CS 5402: Introduction to Data Mining

SP2026

(Lecture -9&10)

Logistic Regression & SVM

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Course Project

- Team: [Link]
- Team & Course Project Registration [TBD: Link]
- Deadline: 02/24/2026

From Perceptron to Logistic Regression

Problem with Perceptron

- Hard decision (sign function)
- No probability output
- Non-smooth loss
- Not robust to noise

Idea of Logistic Regression

- Instead of predicting:

$$\hat{y} = sign(w^T x + b)$$

- We predict:

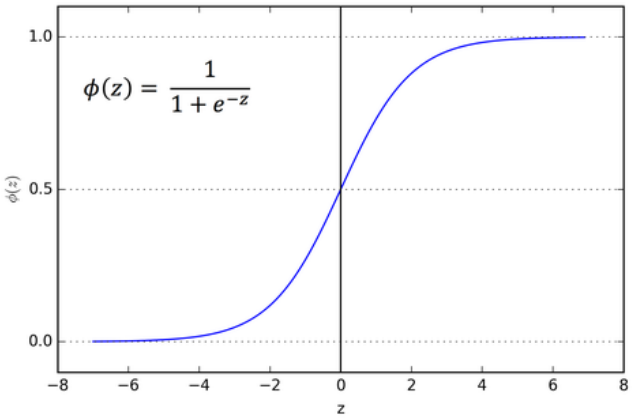
$$P(y = 1 \mid x)$$

- A probability between 0 and 1.

Logistic (Sigmoid) Function

Sigmoid Function

$$\sigma(z) = \frac{1}{1+e^{-z}} \quad \text{Where: } z = w^T x + b$$



Properties:

- Output in (0,1)
- Smooth and differentiable
- Monotonic increasing
- Converts linear score → probability

Prediction Rule

$$P(y = 1 \mid x) = \sigma(w^T x + b)$$

Classify as:

$$\hat{y} = \begin{cases} 1 & \text{if } P \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Loss Function (1)

- For one sample:

$$P(y \mid x) = \sigma(z)^{y}(1-\sigma(z))^{1-y}$$

- For dataset:

$$L(w) = \prod_{i} P\left(y_i \mid x_i\right)$$

- Take log:

$$\ell(w) = \sum_{i}\left[y_i \log \sigma\left(z_i\right) + (1-y_i)\log\left(1-\sigma(z_i)\right)\right]$$

- We maximize log-likelihood
  ⇔ minimize negative log-likelihood

Maximum Likelihood Estimation (MLE)

Loss Function (2)

- Loss function:

$$J(w) = -\sum_i \left[y_i \log \sigma(z_i) + (1 - y_i) \log\left(1 - \sigma(z_i)\right) \right]$$

- Also called:
  - Cross-entropy loss
  - Log loss

Important Property

- This loss is **convex**
  → Global minimum exists
  → Guaranteed convergence with GD

Gradient of Logistic Loss

- For one sample:

$$J_i = -\left[y \log \sigma(z) + (1-y)\log\left(1-\sigma(z)\right)\right]$$

- Derivative:

$$\frac{\partial J_i}{\partial w} = (\sigma(z) - y)x$$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$
$$\sigma'(z) = \sigma(z)\cdot(1-\sigma(z))$$
$$z = w^\top x + b.$$

$$\begin{aligned}
\frac{\partial J}{\partial w} &= -\left[y \cdot \frac{1}{\sigma(z)} \cdot \sigma'(z) \cdot \frac{\partial z}{\partial w} + (1-y)\cdot \frac{1}{1-\sigma(z)} \cdot (-1) \cdot \sigma'(z) \cdot \frac{\partial z}{\partial w}\right] \\
&= -\left[y \cdot \frac{1}{\sigma(z)} \cdot \sigma(z)\cdot(1-\sigma(z)) \cdot x + (1-y)\cdot \frac{-1}{1-\sigma(z)} \cdot \sigma(z)\cdot(1-\sigma(z)) \cdot x\right] \\
&= -[y - y\cdot\sigma(z) - \sigma(z) + y\cdot\sigma(z)]\cdot x \\
&= [\sigma(z) - y]\cdot x
\end{aligned}$$

Gradient Descent Update

- Recall GD rule:

$$w \leftarrow w - \eta \nabla J(w)$$

- Substitute gradient:

$$w \leftarrow w - \eta(\sigma(z) - y)x$$

- Rewritten:

$$w \leftarrow w + \eta\big(y - \sigma(z)\big)x$$

Compare to Perceptron

- Perceptron update:

$$w \leftarrow w + \eta y x \ (\textit{if misclassified})$$

- Logistic update:

$$w \leftarrow w + \eta (y - \hat{p}) x$$

- Smooth and continuous update!

Interpretation of Update

$$w \leftarrow w + \eta\big(y - \sigma(z)\big)x$$

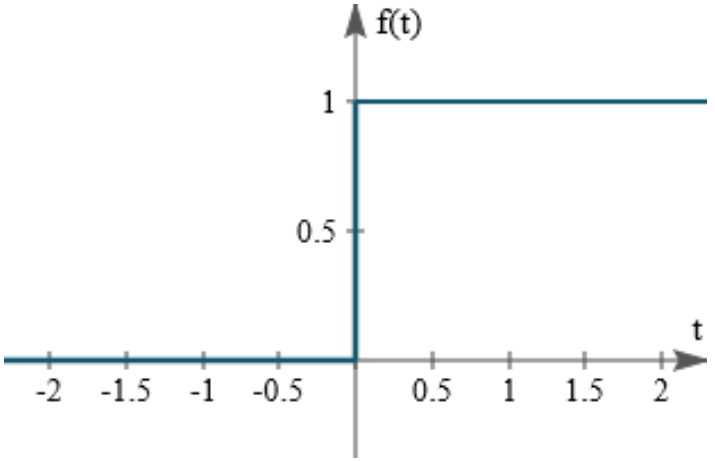
Term: $\big(y - \sigma(z)\big)$ is the error.

If:

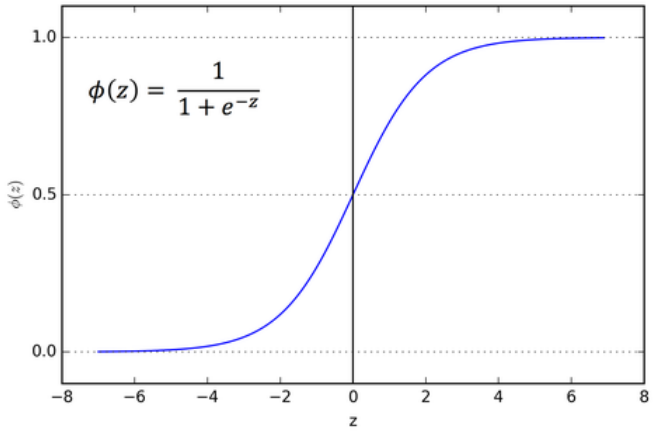
- Model underestimates probability → increase weights
- Model overestimates → decrease weights
- Small error → small update
- This makes logistic regression stable.

Visualization of Difference

- Perceptron:
  - Sharp boundary
  - Step-like behavior

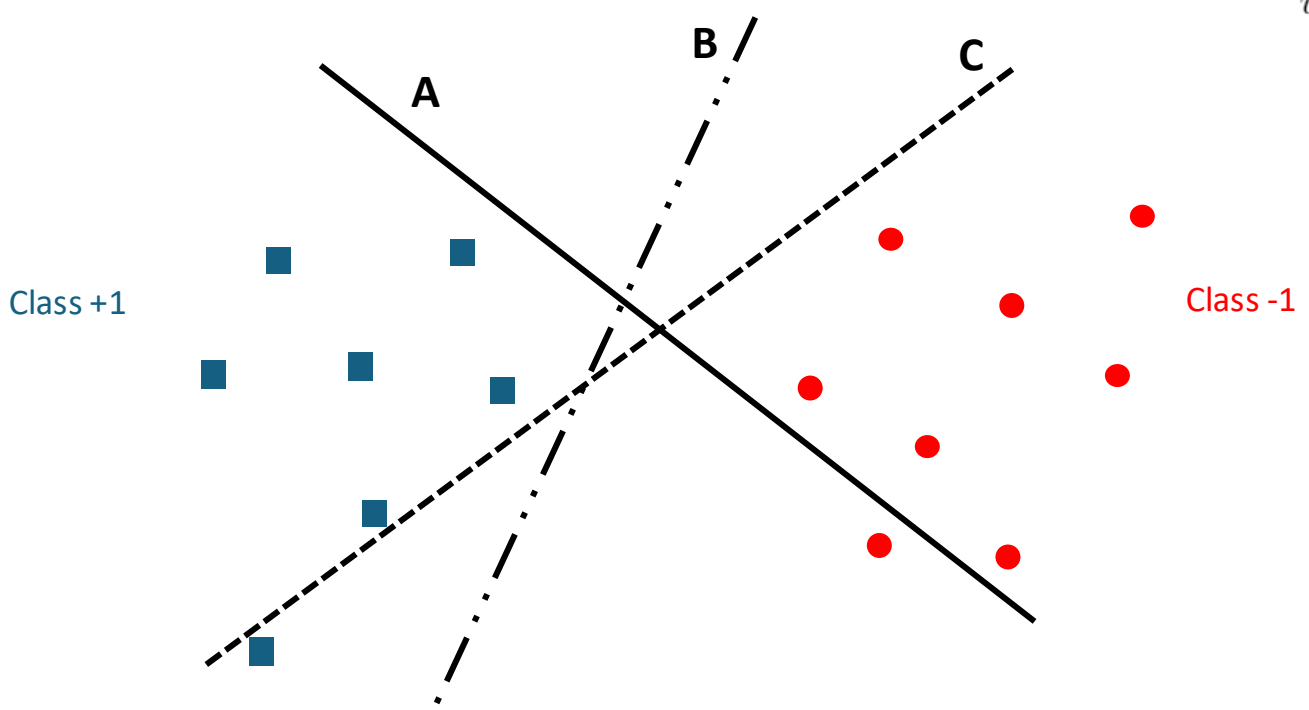


- Logistic Regression:
  - Smooth S-shaped curve
  - Confidence increases gradually

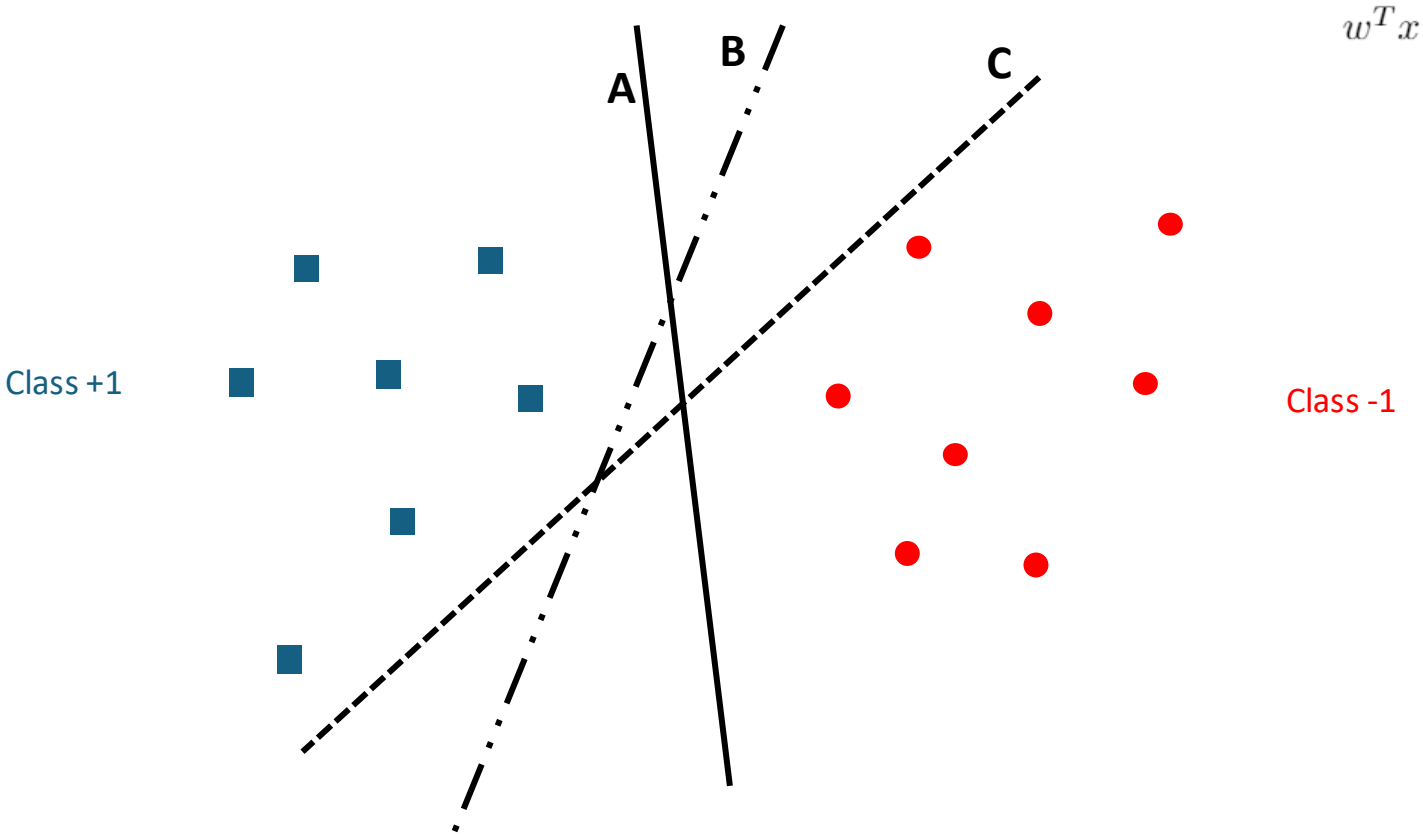


Review: Perceptron (1)

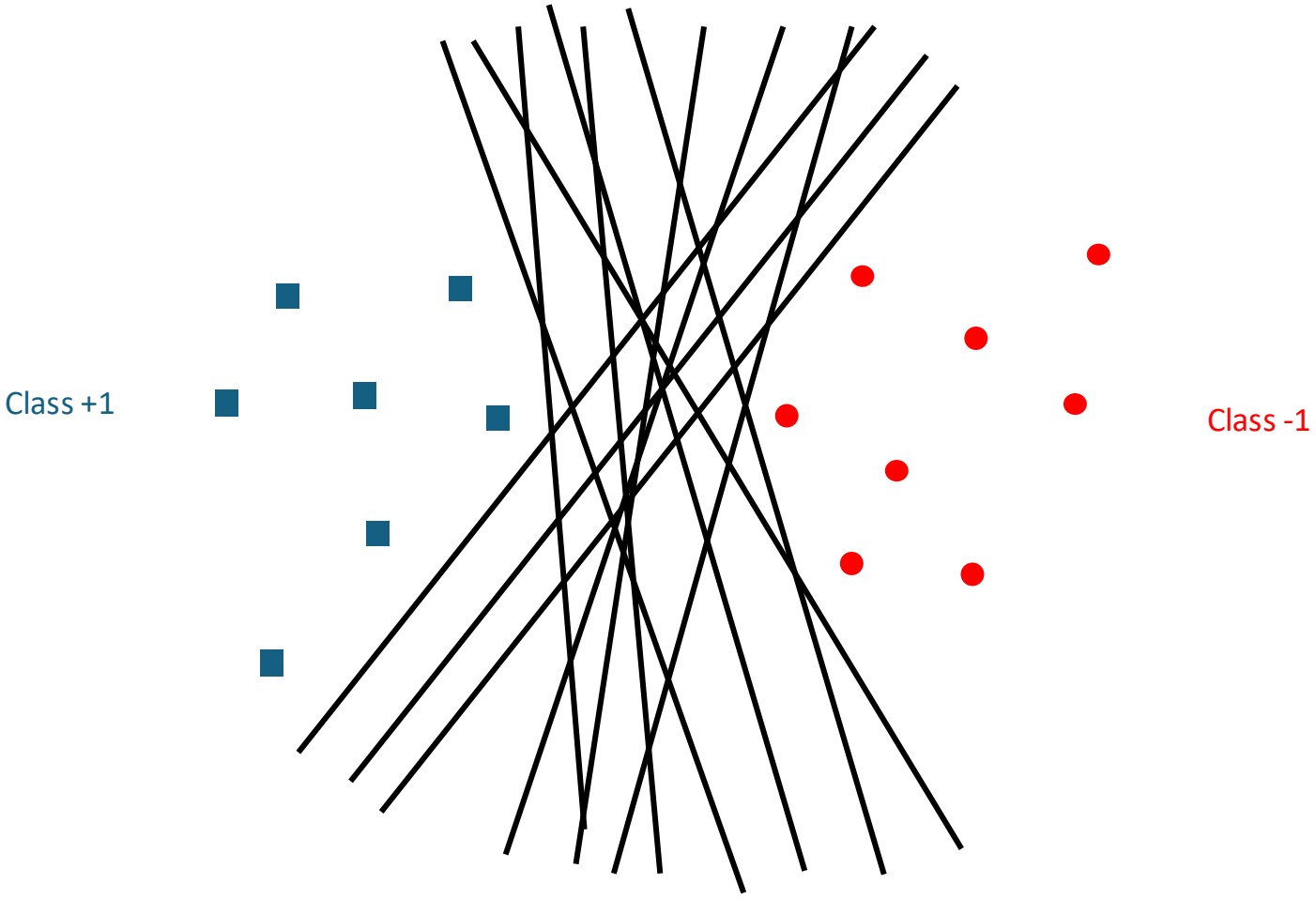
$$w^T x + b = 0$$



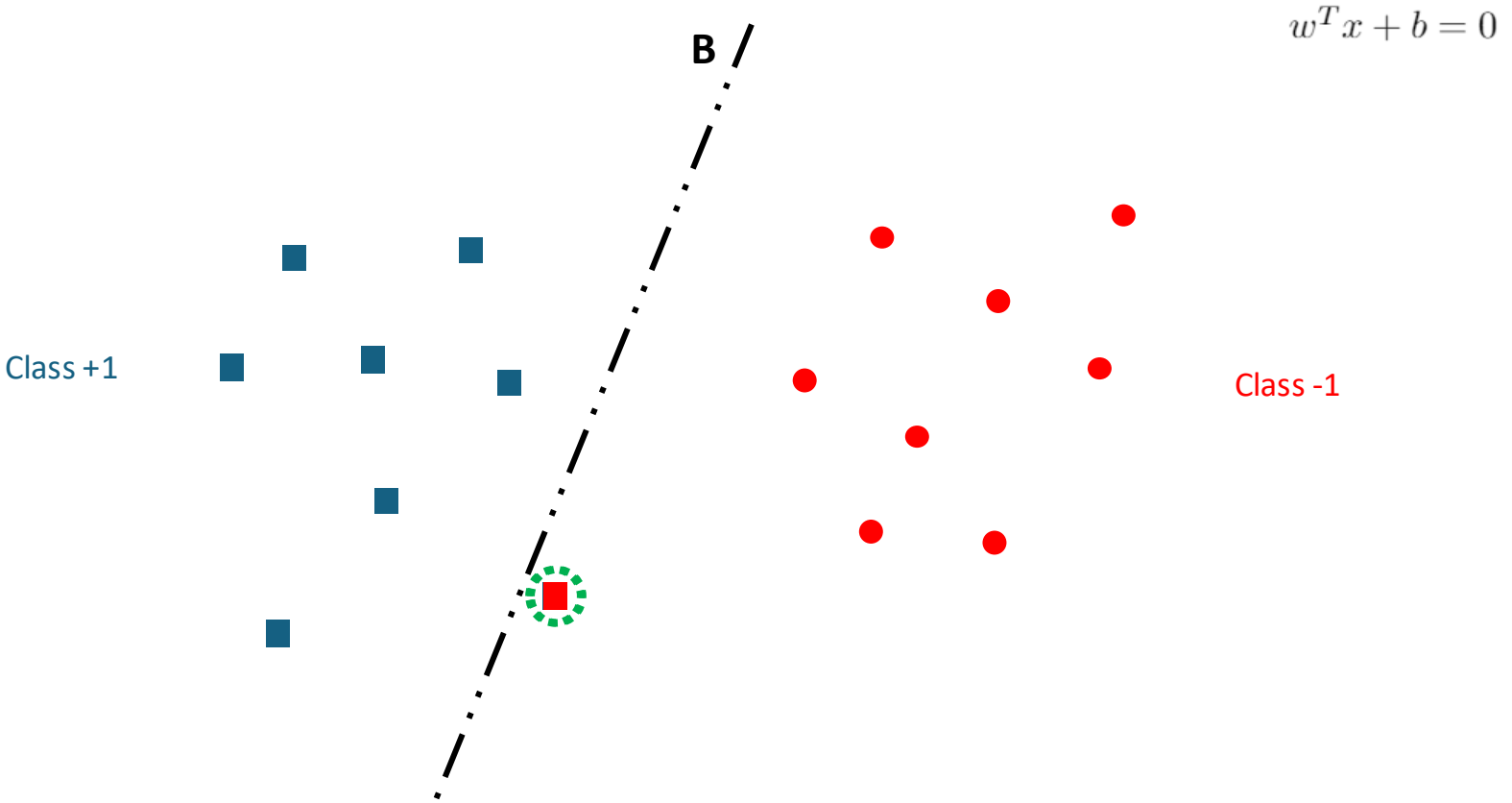
Review: Perceptron (2)



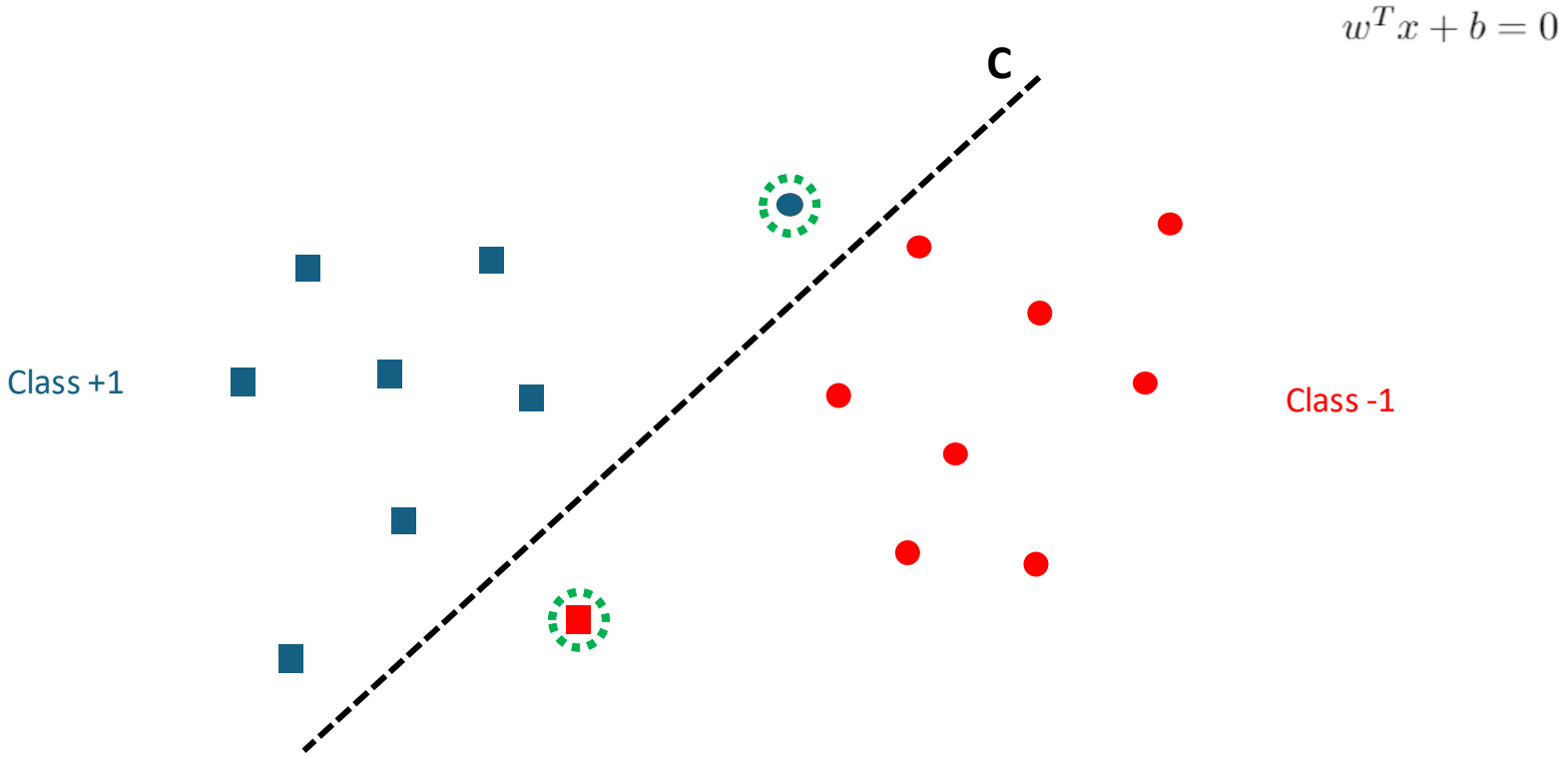
Review: Perceptron (3)



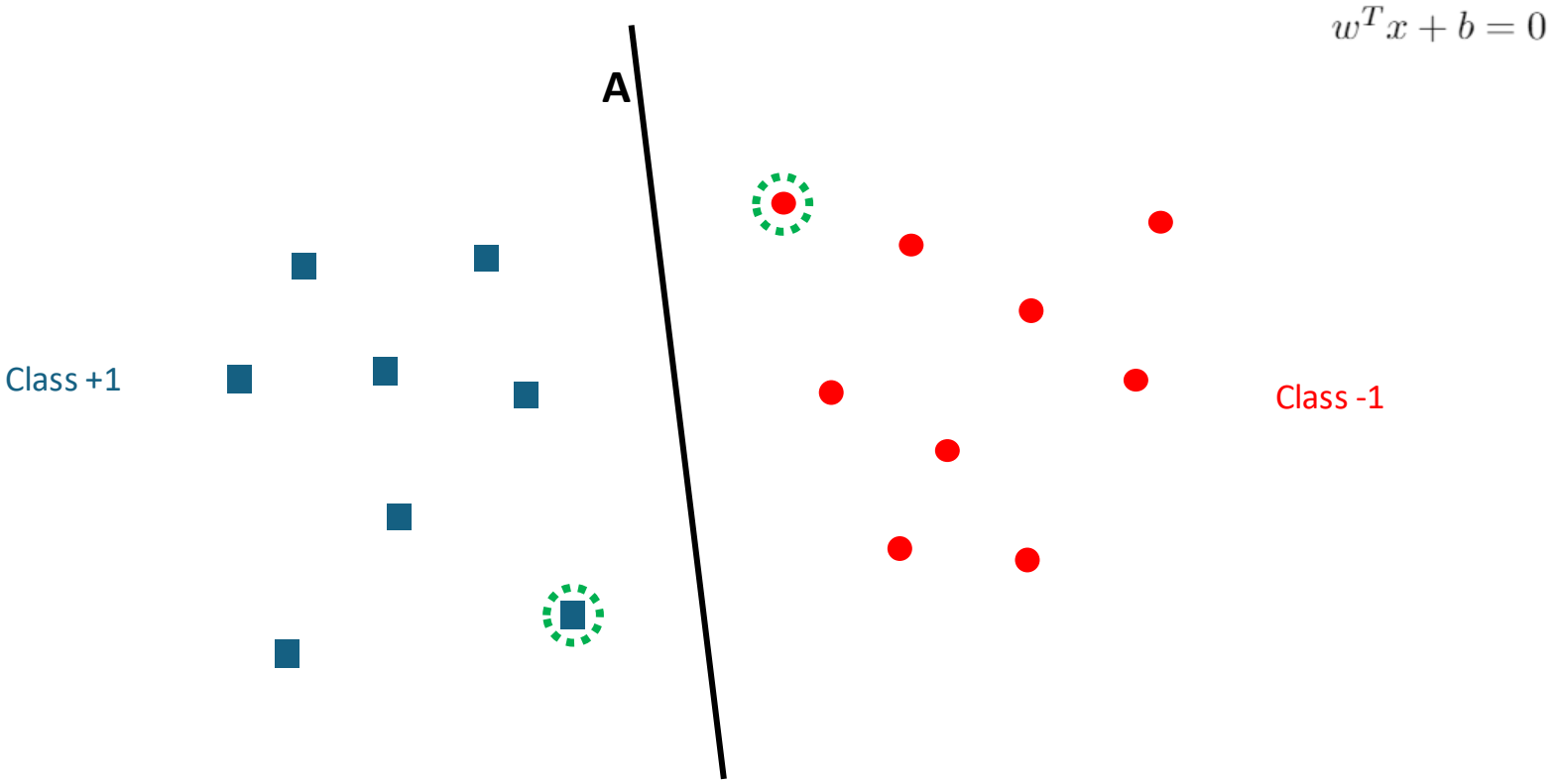
Review: Perceptron (4)



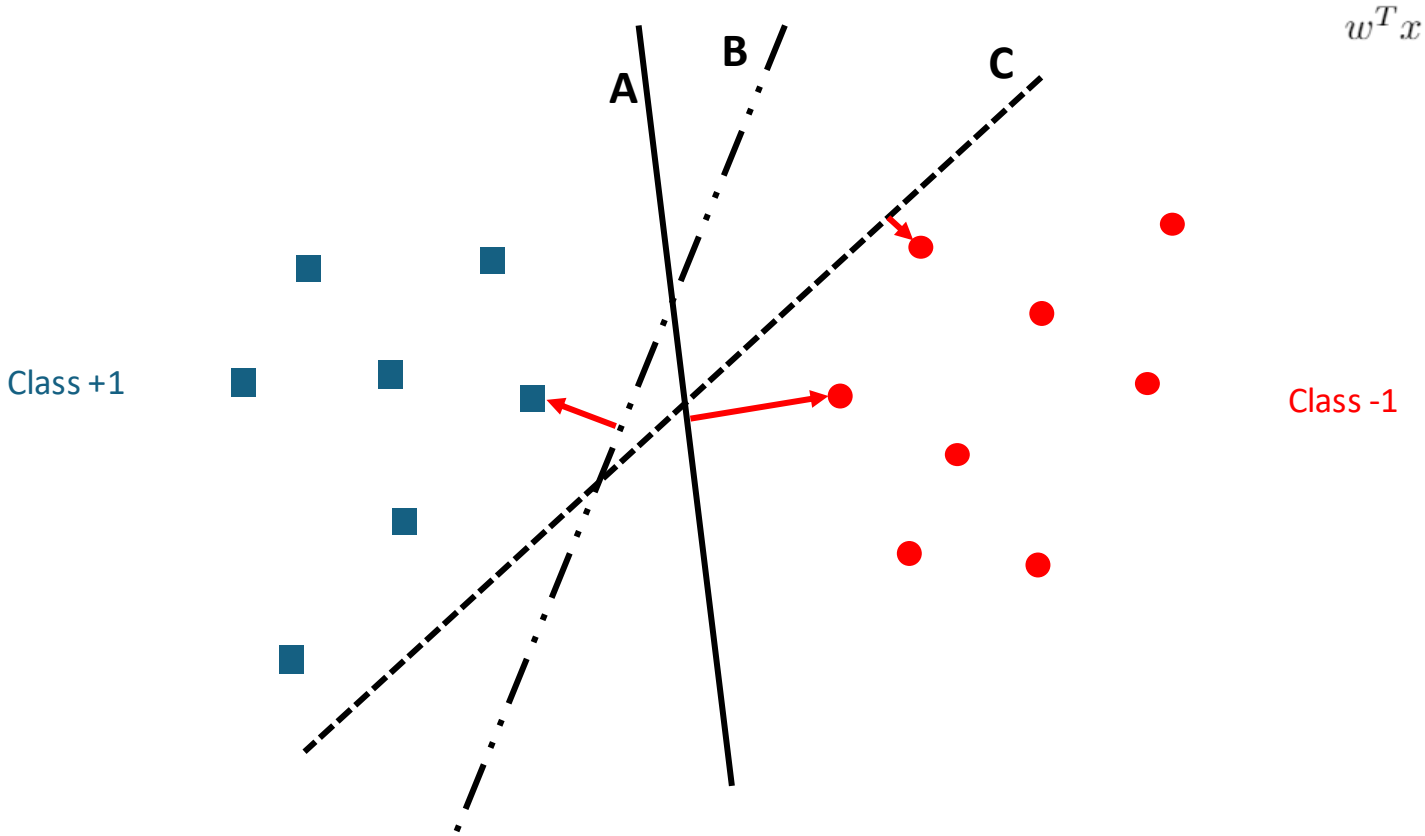
Review: Perceptron (5)



Review: Perceptron (6)



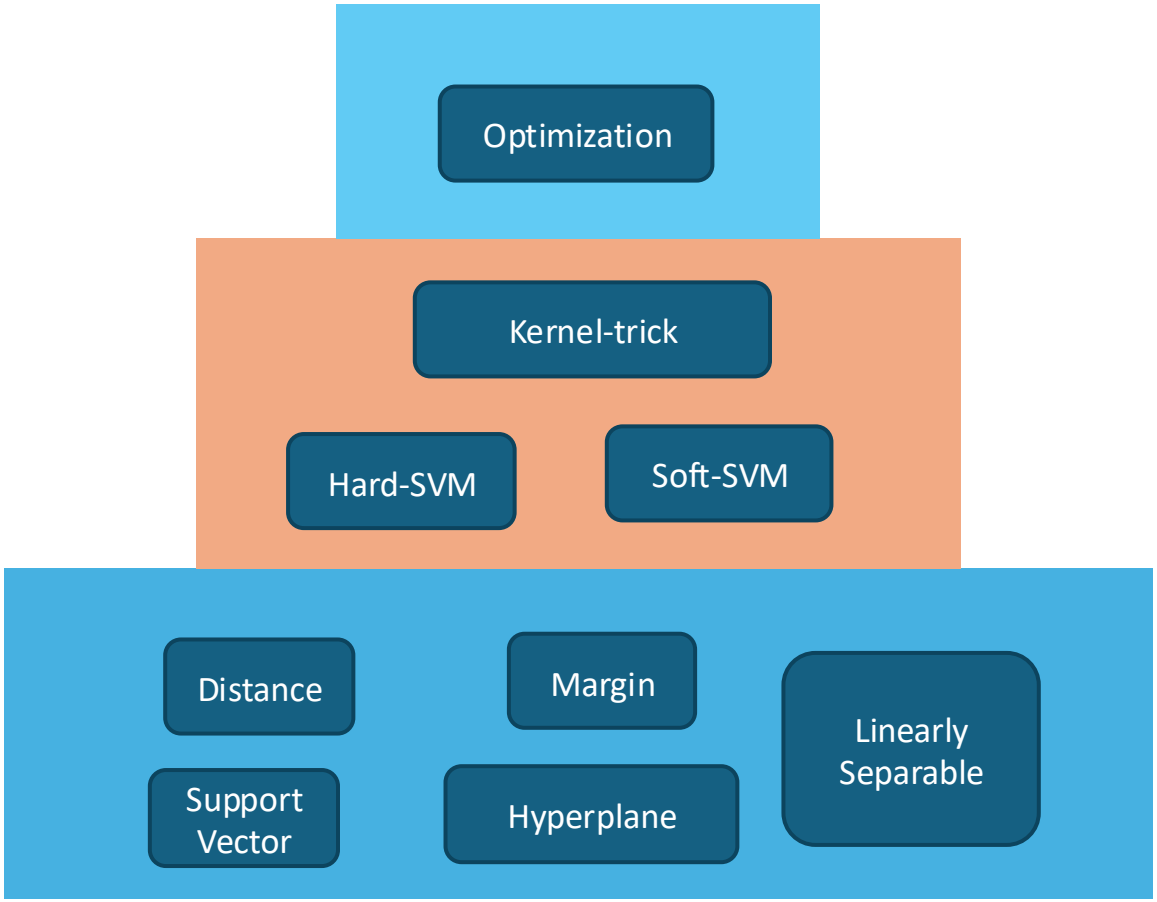
Review: Perceptron (7)



Contents

- Foundation of SVM
  - Margin
  - Support vector
  - Hyperplane
- Advanced topics:
  - Hard-SVM
  - Soft-SVM
  - Optimization
  - Kernel tricks

Three levels in SVM



SVM: First Impression

1. Correctly separate the classes

2. Maximize the “distance”

SVM: Hyperplane

- The same as we used in Perceptron.
- Form of equation defining the decision surface separating the classes is a hyperplane of the form:

$$w^T x + b = 0$$

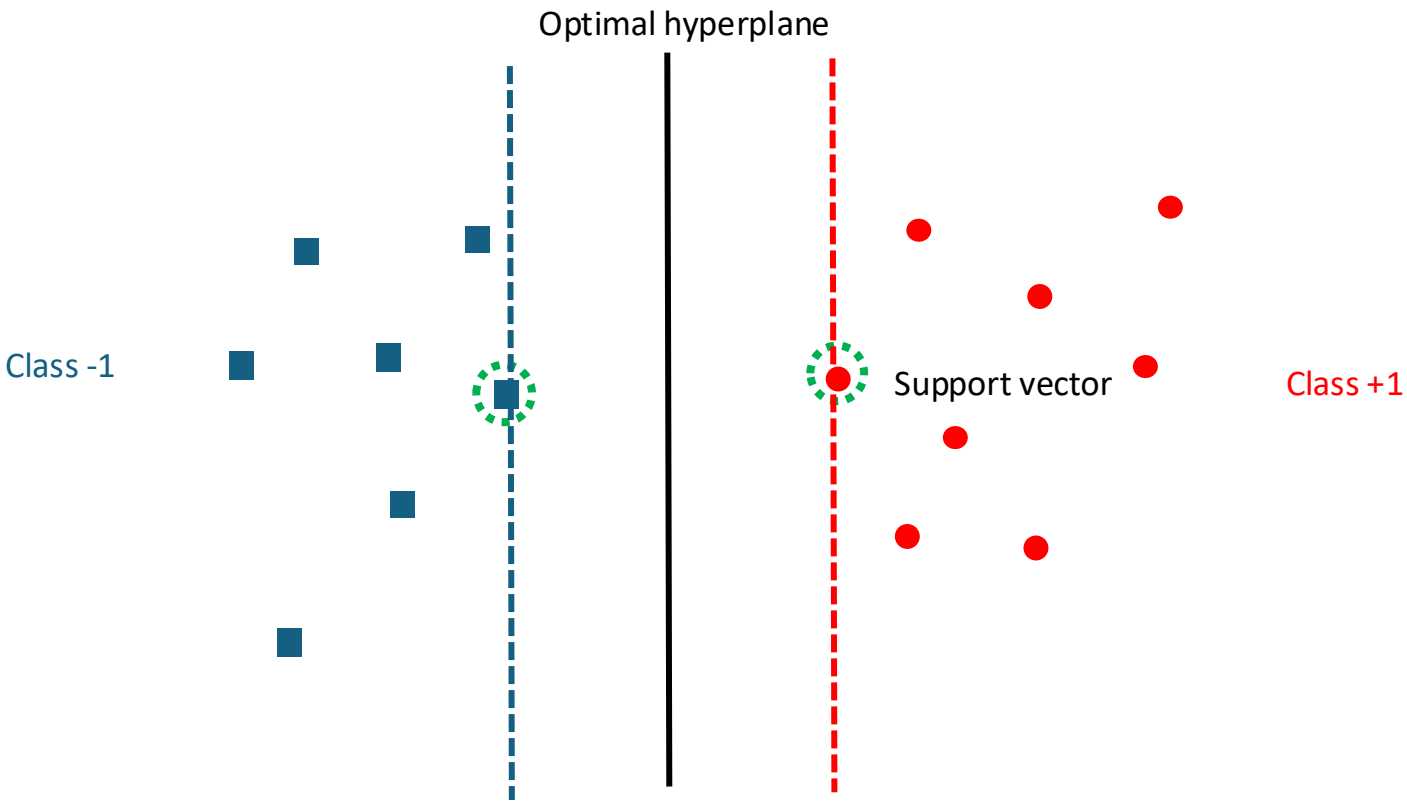
  - W is a weight vector
  - x is input vector
  - b is bias

- Allow us to write:

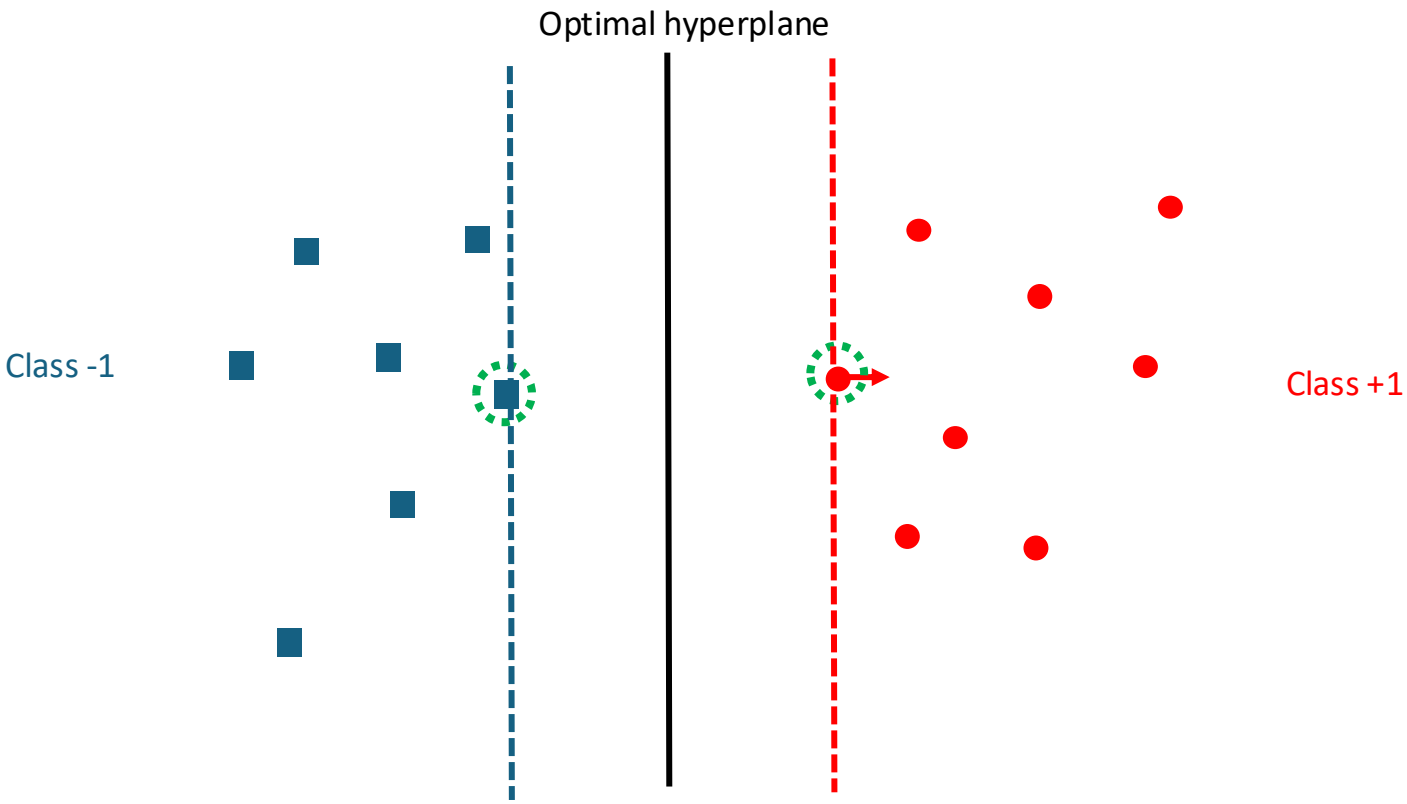
$$w^T x + b > 0 \qquad \text{for +1 class}$$

$$w^T x + b < 0 \qquad \text{for -1 class}$$

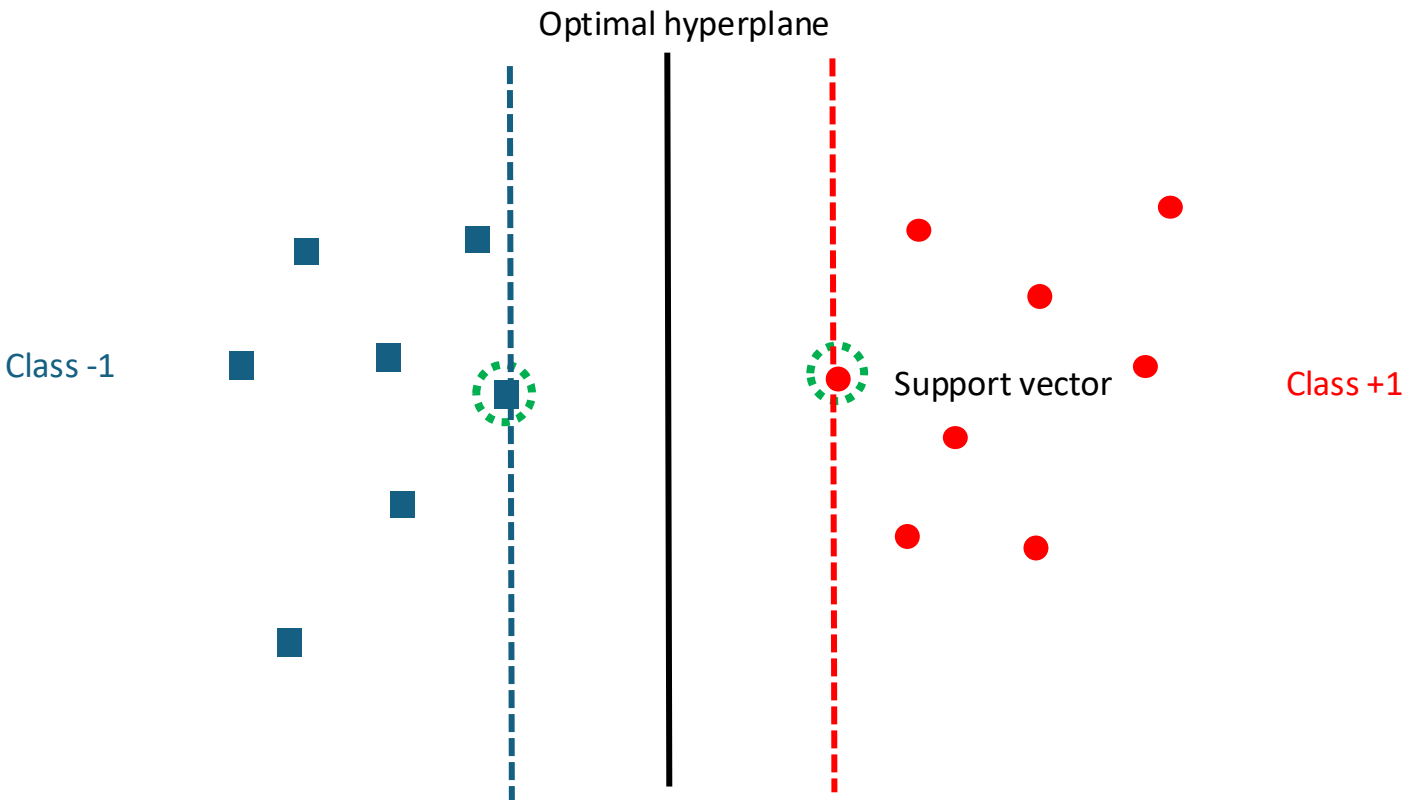
SVM: Support Vector (1)



SVM: Support Vector (2)



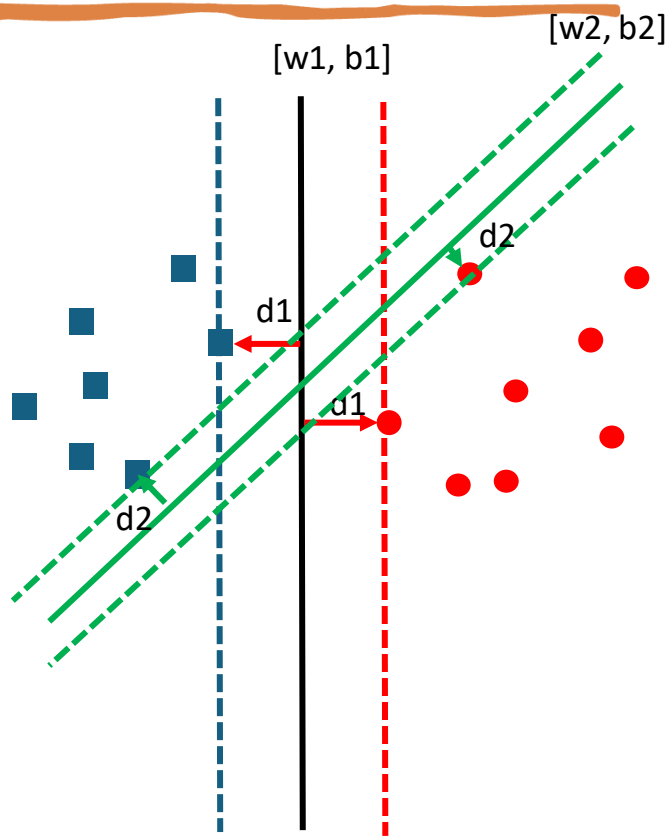
SVM: Support Vector (3)



SVM: margin and hyperplane

- Margin of Separation (d): the separation between the hyperplane and the closet data point for a given weight vector w and bias b.

- Optimal Hyperplane (maximal margin): the particular hyperplane for which the margin of separation d is maximized.



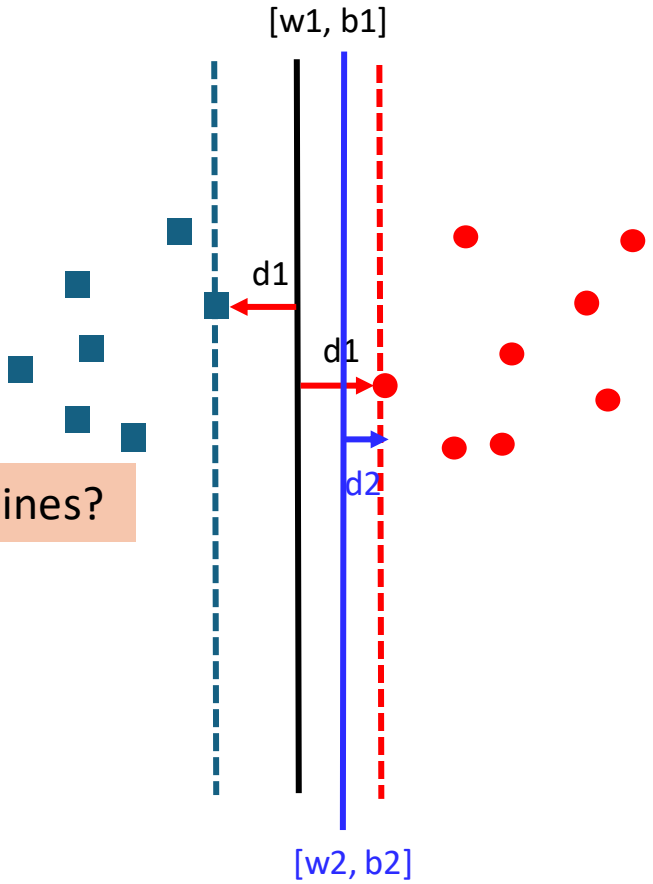
SVM: margin and hyperplane (Cont'd)

- Margin of Separation (d): the separation between the hyperplane and the closet data point for a given weight vector w and bias b.

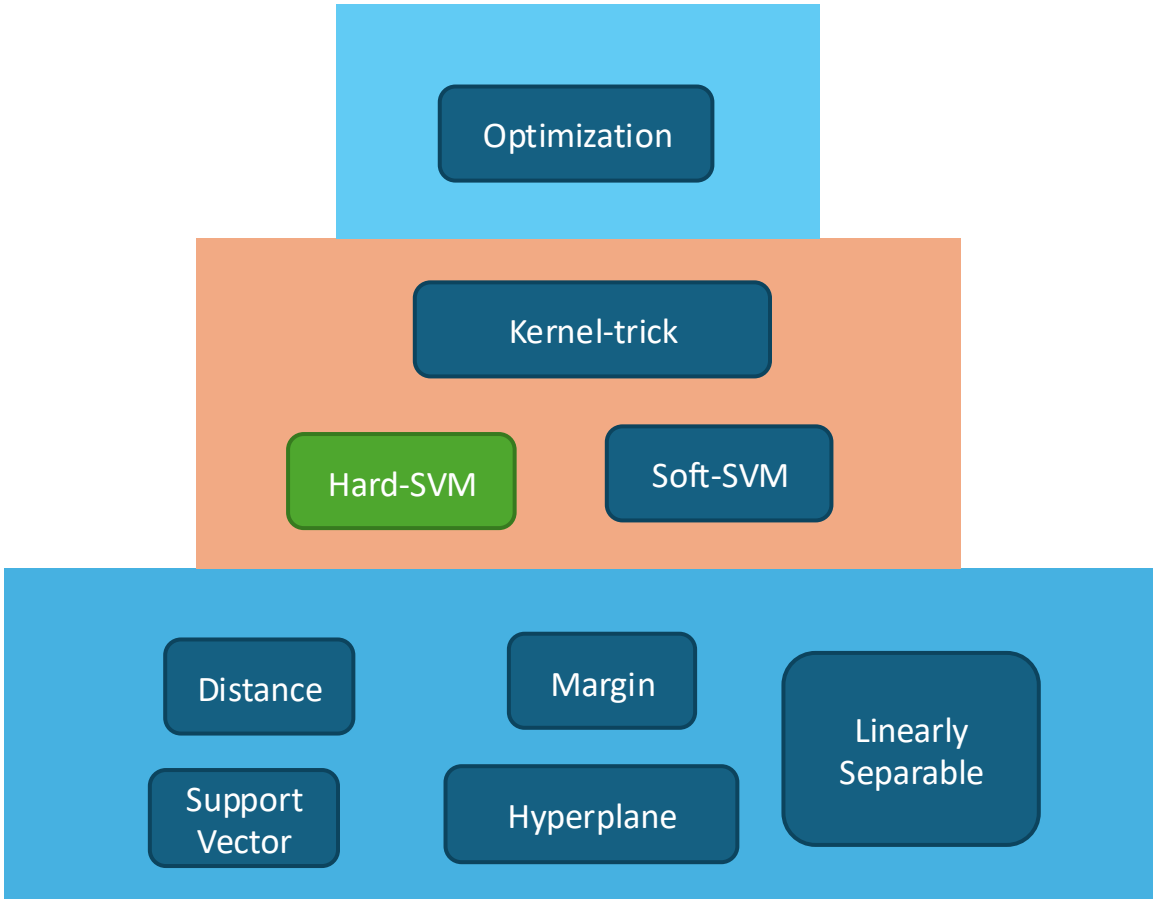
- Optimal Hyperplane (maximal margin): the particular hyperplane for which the margin of separation d is maximized.

Why $d_1 = d_1$ in optimal hyperplane?

Why does the black line lie in the middle of the two dotted lines?



Three levels: Hard-SVM



SVM: objective (1)

Optimal Hyperplane: maximize the margin of separation.

- Margin over all training data points:

$$\gamma = \min_i \frac{|f_{w,b}(x_i)|}{||w||}$$

- Since only want correct $f_{w,b}(x)$, and recall $y_i \in \{-1, +1\}$, we have:

$$\gamma = \min_i \frac{y_i f_{w,b}(x_i)}{||w||}$$

- Why?

SVM: objective (2)

- Maximize margin over all training data points:

γ:

$$\max_{w,b} \gamma = \max_{w,b} \left\{ \min_i \frac{y_i f_{w,b}(x_i)}{||w||} \right\} = \max_{w,b} \left\{ \min_i \frac{y_i (w^T x_i + b)}{||w||} \right\}$$

- A bit complicated…

Question: Geometric margin

Geometric margin:

$$\frac{y_i\left(w^T x_i + b\right)}{\|w\|} \quad \Rightarrow \quad \frac{y_i\left(cw^T x_i + cb\right)}{\|cw\|}$$

If we scale (w, b) by a factor c, what will happen to margin?

Question: fix one quantity

Since the problem is scale-invariant, we must fix one quantity.

Which would you choose?

A. Fix $\| w \| = 1$

B. Fix $y_i(w^T x_i + b) = 1$

$$\frac{y_i \left(w^T x_i + b\right)}{\|w\|}$$

Why?

Fix $\| w \| = 1$

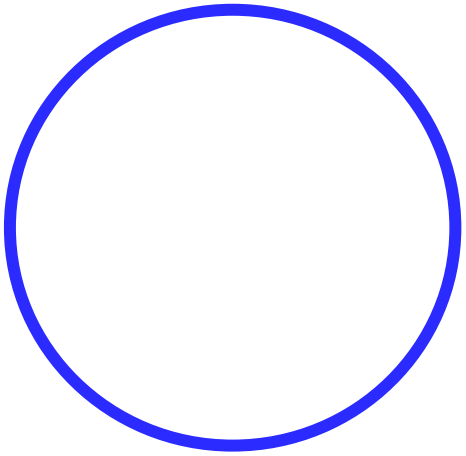
If we set:

$$\| w \| = 1$$

The optimization becomes:

$$\max_{w,b} \min_{i} y_i(w^T x_i + b)$$

subject to: $\| w \| = 1$

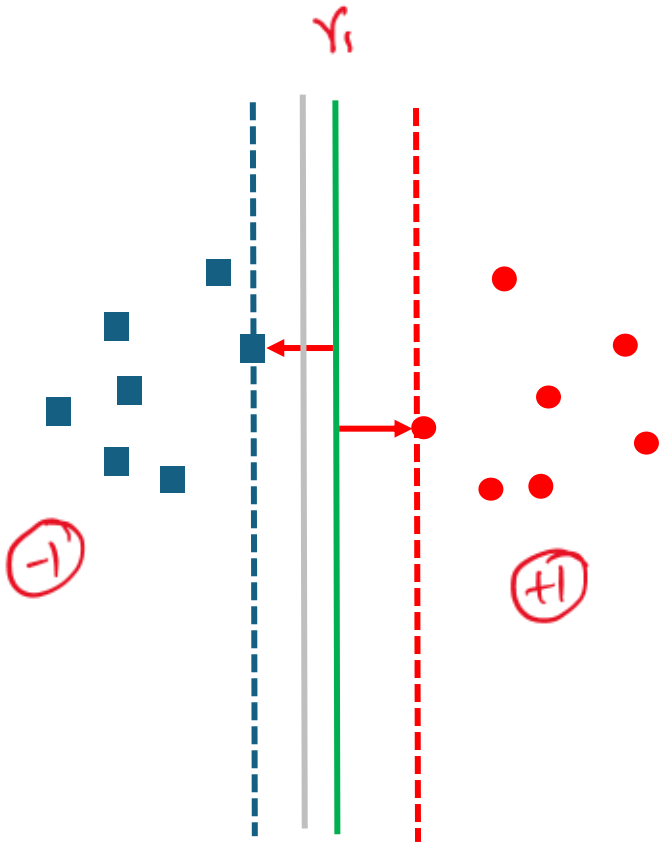


The set: $\{w: \| w \| = 1\}$ is the surface of a sphere.

It is **not a convex set**.

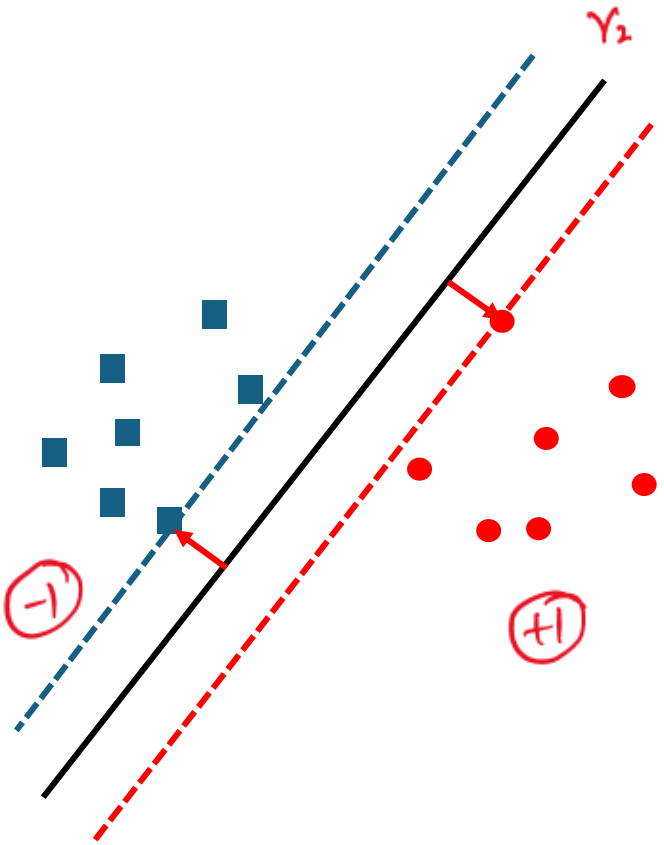
Intuitive Example (1)

Objective: $\max_{w,b} \gamma$



Intuitive Example (2)

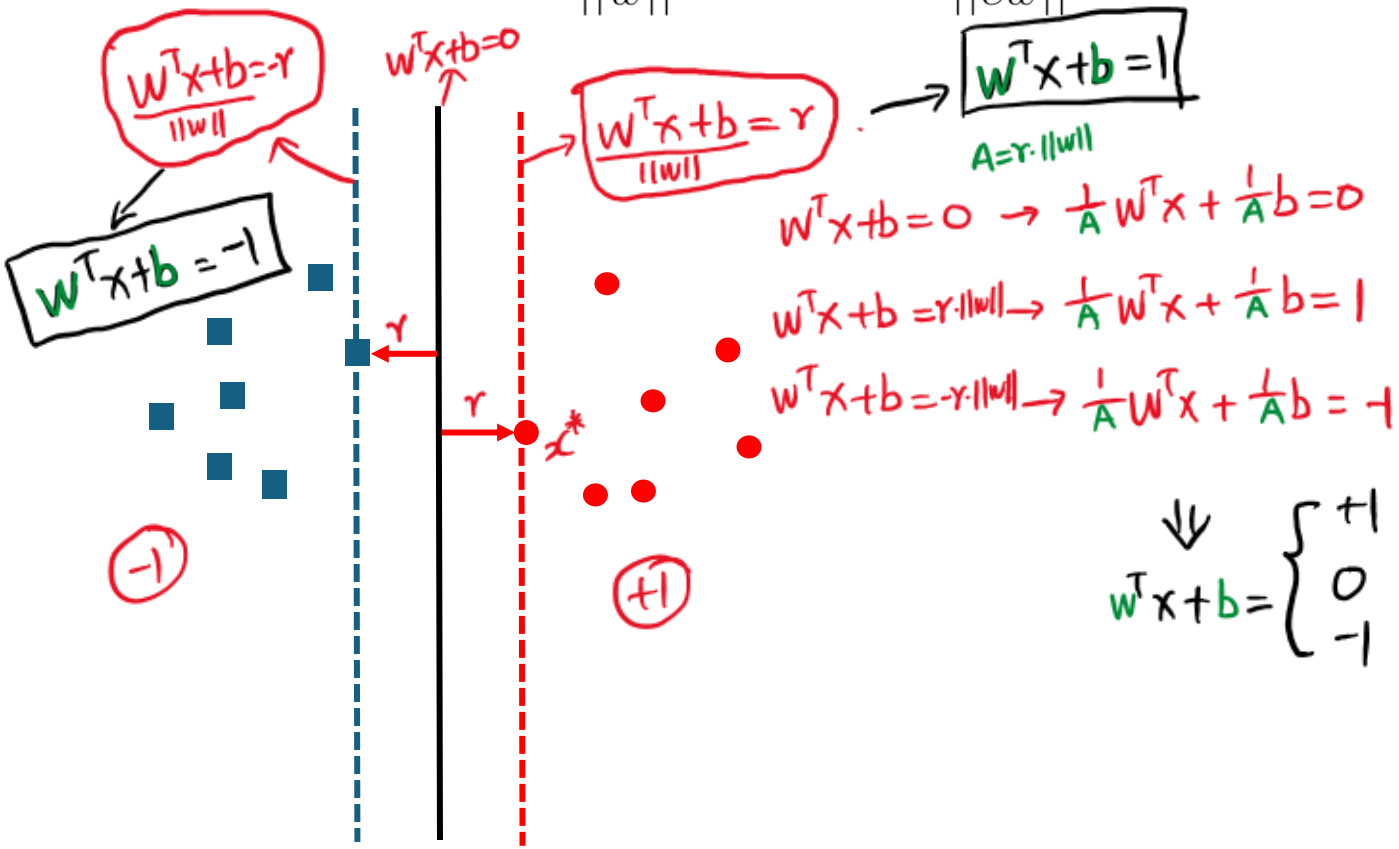
Objective: $\max_{w,b} \gamma$



Intuitive Example (3)

- Observation: when (w, b) scaled by a factor c, the margin unchanged

$$\frac{y_i(w^T x_i + b)}{||w||} = \frac{y_i(cw^T x_i + cb)}{||cw||}$$



Fix $y_i(w^T x_i + b) = 1$

If we set:

$$y_{i*}(w^T x_{i*} + b) = 1$$

Where (x_{i*}, y_{i*}) is the support vector (point closest to the hyperplane), and the geometric margin becomes:

$$\gamma = \frac{1}{\|w\|}$$

The optimization becomes:

$$\max_{w,b} \gamma = \max_{w,b} \frac{1}{||w||}$$

subject to: $y_i(w^T x_i + b) \geq 1$

SVM: Simplified objective

$$\max_{w,b} \left\{ \min_i \frac{y_i(w^T x_i + b)}{||w||} \right\}$$

$$= \max_{w,b} \frac{1}{||w||}$$

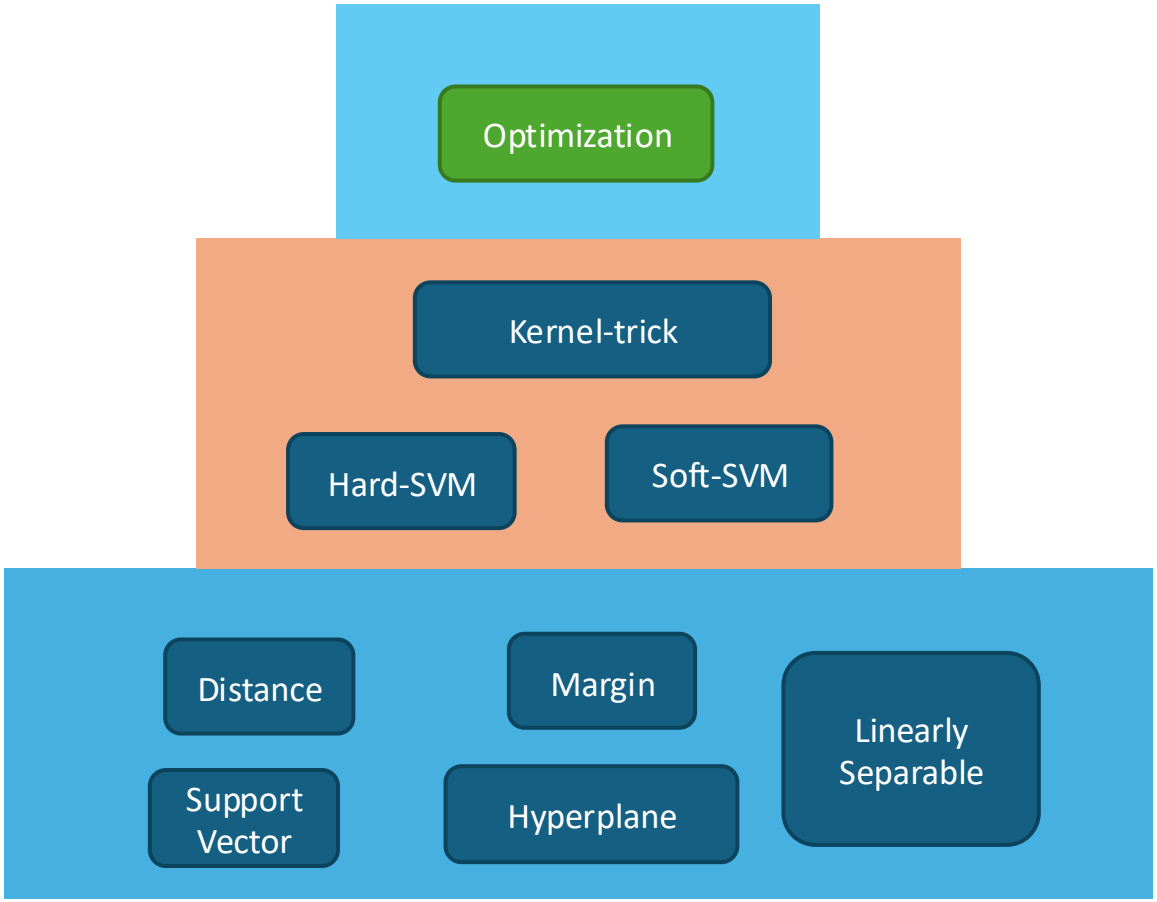
$$= \min_{w,b} \frac{1}{2}||w||^2$$

- Optimization simplified to:

$$\min_{w,b} \frac{1}{2}||w||^2$$
$$s.t.\ y_i(w^T x_i + b) \geq 1$$
$$for\ i\ \in \{1, 2, ..., N\}$$

- Next, how to find the optimum w^* ?

Three levels: Optimization



SVM: Optimization (1)

- Optimization (Quadratic Programming):

$$\min_{w,b} \frac{1}{2}||w||^2$$
$$s.t.\ y_i(w^T x_i + b) \geq 1$$
$$for\ i\ \in \{1, 2, ..., N\}$$

- Solved by Lagrange multiplier method:

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2}||w||^2 + \sum_i \alpha_i [1 - y_i(w^T x_i + b)]$$

Were $\alpha_i \geq 0$ is the Lagrange multiplier.

Lagrange and Dual Problem

- Consider optimization problem:

$$\min_{w} \; f(w)$$

$$\text{s.t. } h_i(w) = 0, \;\; i \in \{1, 2, ..., l\}$$

Any feasible w, a primal value.

Min f(w*) = Optimal primal value.

$$p^* = \min_{w \text{ feasible}} f(w)$$

- Lagrangian:

$$\mathcal{L}(w, \beta) = f(w) + \sum_i \beta_i h_i(w)$$

$$\beta_i \geq 0$$

$$q(\beta) \leq f(w)$$

dual value ≤ primal value

Strong duality: $d^* = p^*$

- Dual problem:

$$q(\beta) = \inf_x \mathcal{L}(\mathrm{x}, \beta)$$

$$\max_{\beta \geq 0} q(\beta)$$

For any $\beta \geq 0$, a dual value;

Max q(β) = optimal dual value

$$d^* = \max_{\beta \geq 0} q(\beta)$$

$$\inf_x \mathcal{L}(x, \beta) \approx \min_x \mathcal{L}(x, \beta)$$

Review: Lagrange

- Consider optimization problem:

$$\min_{w} \; f(w)$$

$$\text{s.t.} \;\; h_i(w) = 0, \;\; i \in \{1, 2, ..., l\}$$

- Solved by setting derivatives of Lagrangian to 0:

$$\mathcal{L}(w, \beta) = f(w) + \sum_i \beta_i h_i(w)$$

$$\frac{\partial \mathcal{L}}{\partial w_i} = 0; \;\; \frac{\partial \mathcal{L}}{\partial \beta_i} = 0$$

Example: Lagrange

- Examples:

$$argmin\ f(x,y) = x + y$$
$$s.t.\ \ x^2 + y^2 = 1$$

$$L(x, y, \beta) = x + y + \beta(x^2 + y^2 - 1)$$

$$\frac{\partial L}{\partial x} = 1 + 2\beta \cdot x = 0 \Rightarrow x = -\frac{1}{2\beta}$$

$$\frac{\partial L}{\partial y} = 1 + 2\beta \cdot y = 0 \Rightarrow y = -\frac{1}{2\beta}$$

$$\frac{\partial L}{\partial \beta} = x^2 + y^2 - 1 = 0 \Rightarrow \frac{1}{4\beta^2} + \frac{1}{4\beta^2} - 1 = 0$$

$$\Rightarrow \beta^2 = \frac{1}{2}$$

$$\beta = \pm\frac{\sqrt{2}}{2}$$

if $\beta = \frac{\sqrt{2}}{2}$ $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ $-\sqrt{2}$

if $\beta = -\frac{\sqrt{2}}{2}$

Review: Generalized Lagrange

- Consider optimization problem:

$$\min_w \ f(w)$$

$$\text{s.t. } g_i(w) \leq 0, \ i \in \{1, 2, ..., k\}$$

$$h_j(w) = 0, \ j \in \{1, 2, ..., l\}$$

- Generalized Lagrangian:

$$\mathcal{L}(w, \alpha, \beta) = f(w) + \sum_i \alpha_i g_i(w) + \sum_j \beta_j h_j(w)$$

SVM: Optimization (2)

- Optimization (Quadratic Programming):

$$\min_{w,b} \frac{1}{2}||w||^2$$
$$s.t.\ y_i(w^T x_i + b) \geq 1$$
$$for\ i\ \in \{1, 2, ..., N\}$$

- Generalized Lagrangian:

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2}||w||^2 + \sum_i \alpha_i[1 - y_i(w^T x_i + b)]$$

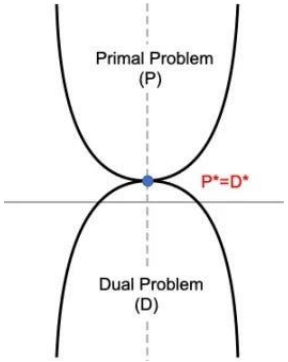
Were α is the Lagrange multiplier.

SVM: Optimization (3)

- Based on the Duality Theory of Constrained Optimization, the dual problem of the primal problem is:

$$\max_{\alpha} \min_{w,b} \mathcal{L}(w, b, \alpha)$$

If we can solve for α (dual problem), Then we have a solution for w, b (primal problem)



Step-1: $\min_{w,b} \mathcal{L}(w, b, \alpha)$

- Calculate the partial gradient, and set as 0:

$$\frac{\partial \mathcal{L}}{\partial w} = 0 \Rightarrow w = \sum_i \alpha_i y_i x_i \qquad \frac{\partial \mathcal{L}}{\partial b} = 0 \Rightarrow 0 = \sum_i \alpha_i y_i$$

- Plug into: $\mathcal{L}(w, b, \alpha) = \frac{1}{2}||w||^2 + \sum_i \alpha_i [1 - y_i(w^T x_i + b)]$

SVM: Optimization (4)

$$\frac{\partial \mathcal{L}}{\partial w} = 0 \Rightarrow w = \sum_i \alpha_i y_i x_i \qquad \frac{\partial \mathcal{L}}{\partial b} = 0 \Rightarrow 0 = \sum_i \alpha_i y_i$$

- Plug into: $\mathcal{L}(w, b, \alpha) = \frac{1}{2}||w||^2 + \sum_i \alpha_i [1 - y_i(w^T x_i + b)]$

$= \frac{1}{2}\left(\sum_i \alpha_i y_i x_i\right)^2 + \sum_i \alpha_i - \sum_i \alpha_i y_i \left(\sum_j \alpha_j y_j x_j\right)^T x_i - \sum_i \alpha_i y_i b$

$= 0$

$= \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j \qquad = \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j$

$= \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j .$

$$\mathcal{L}(w, b, \alpha) = \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

Eliminates explicit parameters w

SVM: Optimization (5)

- Based on the Duality Theory of Constrained Optimization, the dual problem of the primal problem is:

$$\max_{\alpha} \min_{w,b} \mathcal{L}(w, b, \alpha)$$

Step-2: $\max_{\alpha} \mathcal{L}(w, b, \alpha)$

- Calculate the α^* for the dual problem:

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

$$\text{s.t.} \sum_i \alpha_i y_i = 0,$$
$$\alpha_i \geq 0$$

SVM: Optimization (6)

- Based on the Duality Theory of Constrained Optimization, the dual problem of the primal problem is:

$$\max_{\alpha} \min_{w,b} \mathcal{L}(w, b, \alpha)$$

Step-3: w, b

- Calculate w, b based on the α^* from the dual problem:

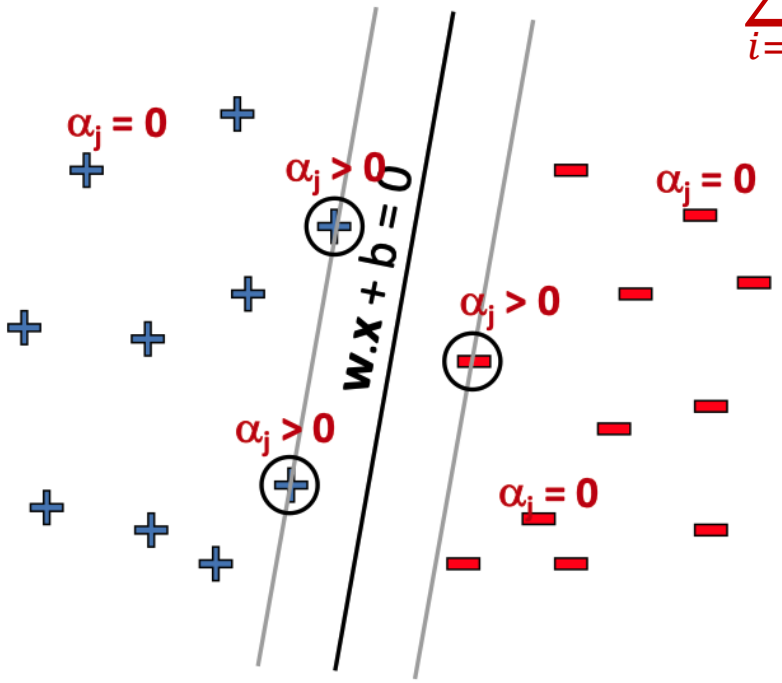
$$w^* = \sum_{i=1}^{N} \alpha_i y_i x_i$$

SVM: Optimization (7)

Step-3: w, b

- Calculate w, b based on the α^* from the dual problem:

$$w^* = \sum_{i=1}^{N} \alpha_i y_i x_i$$



Only few a can be non-zero: Where constraint is active and tight:

$$(w^T x_i + b) y_i = 1$$

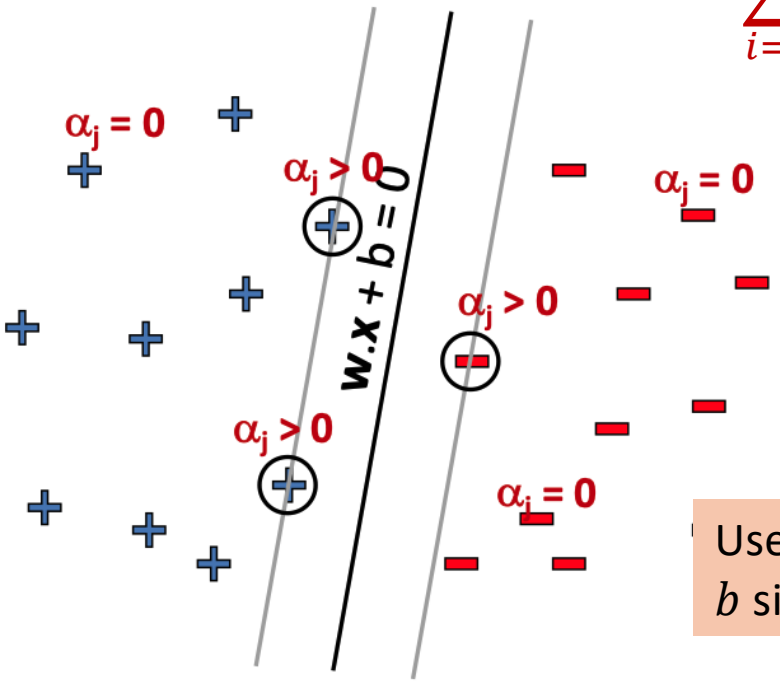
Support vectors- training points i whose α_i are non-zero.

SVM: Optimization (8)

Step-3: w, b

- Calculate w, b based on the α^* from the dual problem:

$$w^* = \sum_{i=1}^{N} \alpha_i y_i x_i$$



Only few a can be non-zero: Where constraint is active and tight:

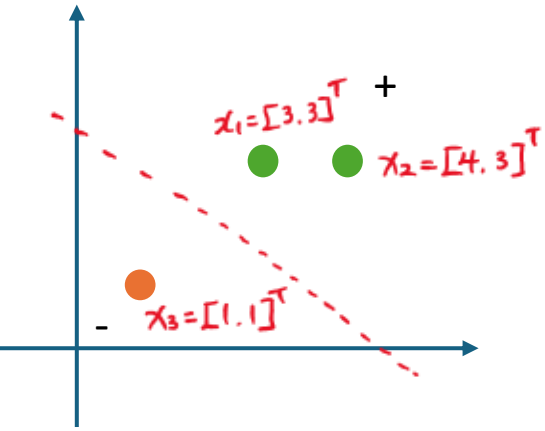
$$(w^T x_i + b) y_i = 1$$

Support vectors- training points i whose α_i are non-zero.

Use any one of support vectors with $\alpha_i > 0$ to compute b since constraint is tight: $(w^T x_i + b) y_i = 1$

$$b = \frac{1}{y_k} - w^T x_k \text{ for any k where } \alpha_k > 0$$

SVM: Example (1)



$$\min_{\alpha} \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_i\alpha_j \cdot y_i \cdot y_j \cdot x_i^T \cdot x_j - \sum_{i=1}^{N}\alpha_i$$

$$s.t. \quad \sum_{i=1}^{N}\alpha_i \cdot y_i = 0.$$

$$\alpha_i \geq 0. \qquad i = 1, 2, \cdots N.$$

$N=3$ in this example:

$$\Rightarrow \min_{\alpha} \frac{1}{2}(18\alpha_1^2 + 25\alpha_2^2 + 2\alpha_3^2 + 42\alpha_1\alpha_2 - 12\alpha_1\alpha_3 - 14\alpha_2\alpha_3) - (\alpha_1 + \alpha_2 + \alpha_3)$$

$$s.t: \quad \alpha_1 + \alpha_2 - \alpha_3 = 0.$$

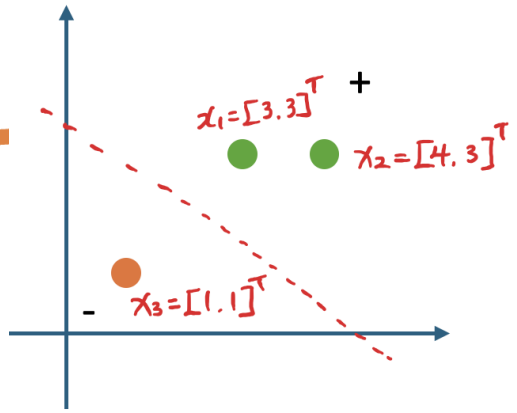
$$\alpha_1 \geq 0. \quad \alpha_2 \geq 0. \quad \alpha_3 \geq 0.$$

SVM: Example (2)

$$\Rightarrow \min_{\alpha} \frac{1}{2}(18\alpha_1^2 + 25\alpha_2^2 + 2\alpha_3^2 + 42\alpha_1\alpha_2 - 12\alpha_1\alpha_3 - 14\alpha_2\alpha_3) - (\alpha_1 + \alpha_2 + \alpha_3)$$

s.t. $\alpha_1 + \alpha_2 - \alpha_3 = 0$.

$\alpha_1 \geq 0$. $\alpha_2 \geq 0$. $\alpha_3 \geq 0$.



plug: $\alpha_3 = \alpha_1 + \alpha_2$

$$S(\alpha_1, \alpha_2) = 4\alpha_1^2 + \frac{13}{2}\alpha_2^2 + 10\alpha_1\alpha_2 - 2\alpha_1 - 2\alpha_2$$

$$\frac{\partial S}{\partial \alpha_1} = 0 \quad \frac{\partial S}{\partial \alpha_2} = 0 \Rightarrow \alpha_1 = \frac{1}{4}, \ \alpha_2 = 0.$$

$$\alpha_3 = \alpha_1 + \alpha_2 = \frac{1}{4}.$$

$$\Rightarrow w^* = \sum_{i=1}^{N} \alpha_i \cdot y_i \cdot x_i = \frac{1}{4} \times 1 \times \begin{bmatrix} 3 \\ 3 \end{bmatrix} + \frac{1}{4} \times (-1) \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$b^* = y_1 - (w^*)^T \cdot x_1 = -2$$

$$\Rightarrow f_{w,b}(x) = \frac{1}{2}x^{(1)} + \frac{1}{2}x^{(2)} - 2$$

SVM: Example (3)

$\frac{\partial S}{\partial \alpha_1} = 0$. $\frac{\partial S}{\partial \alpha_2} = 0$ $\Rightarrow$ $\alpha_1 = \frac{1}{4}$. $\alpha_2 = 0$.

$$\left.\begin{aligned} 8\alpha_1 + 10\alpha_2 - 2 = 0 \\ 13\alpha_2 + 10\alpha_1 - 2 = 0 \end{aligned}\right\} \Rightarrow \boxed{\alpha_1 = \frac{3}{2} . \ \alpha_2 = -1 .}$$

(X) $[\alpha_i \geq 0.]$

$\Rightarrow$ the minimum value should be on the boundary.

$$S(\alpha_1, \alpha_2) = 4\alpha_1^2 + \frac{13}{2}\alpha_2^2 + 10\alpha_1\alpha_2 - 2\alpha_1 - 2\alpha_2$$

$\alpha_1 = 0$ OR $\alpha_2 = 0$.

$S(\alpha_1, \alpha_2) = \frac{13}{2}\alpha_2^2 - 2\alpha_2$

$\frac{\partial S}{\partial \alpha_2} = 0 \Rightarrow \alpha_2 = \frac{2}{13}$.

$S(\alpha_1, \alpha_2) = -\frac{2}{13}$.

$>$

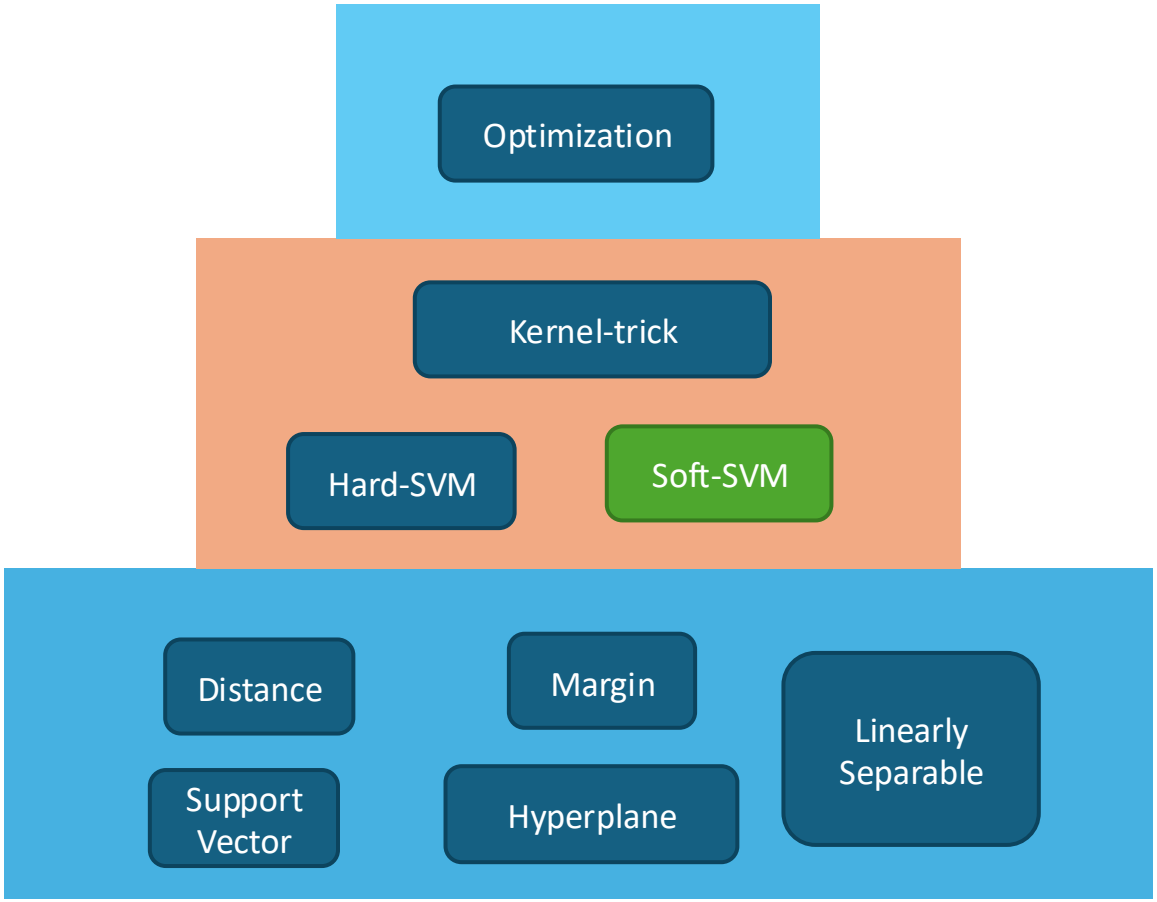
$S(\alpha_1, \alpha_2) = 4\alpha_1^2 - 2\alpha_1$

$\frac{\partial S}{\partial \alpha_1} = 0 \Rightarrow \alpha_1 = \frac{1}{4}$

$S(\alpha_1, \alpha_2) = -\frac{1}{4}$. (✓)

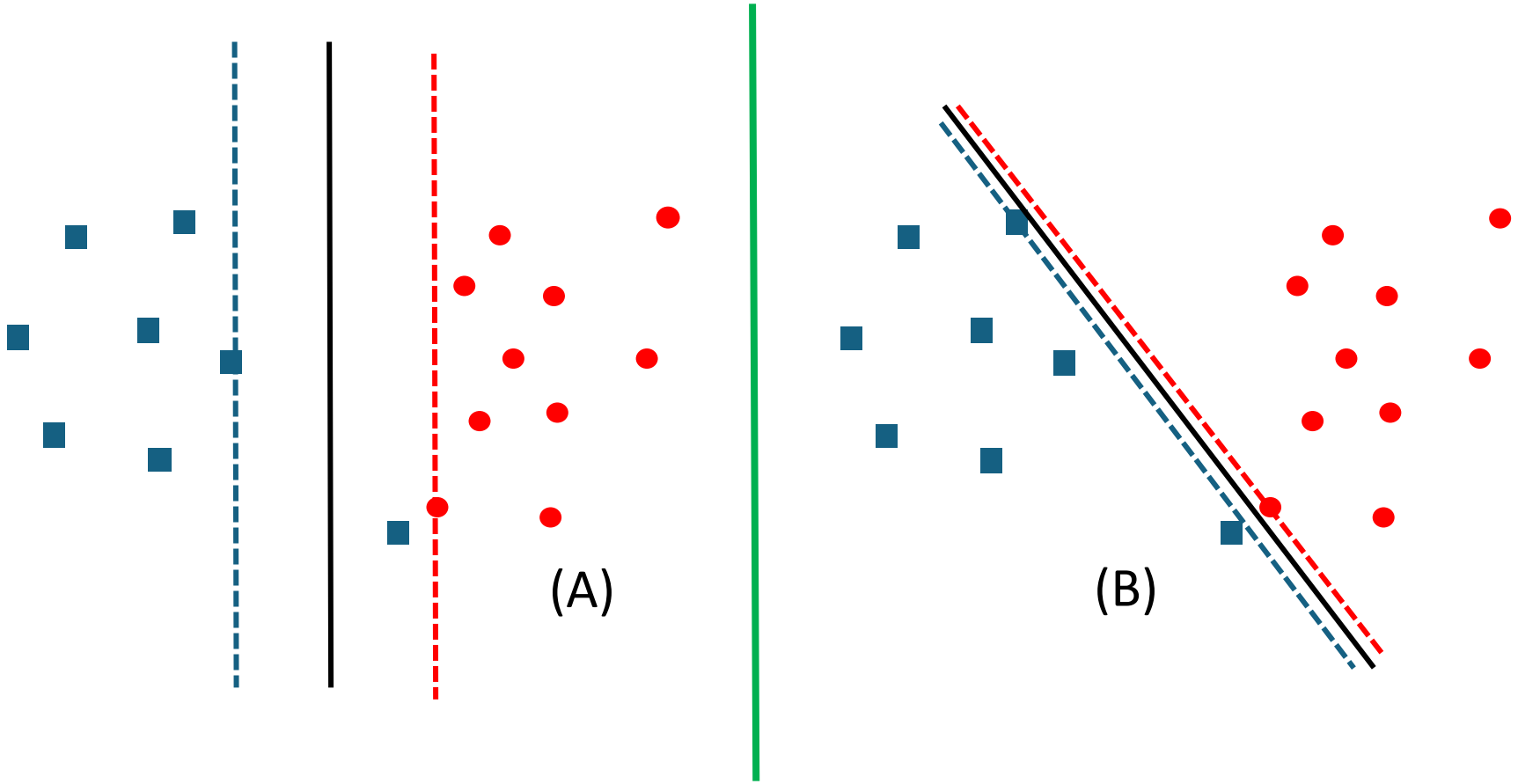
$\alpha_1 = \frac{1}{4}$. $\alpha_2 = 0$.

Three levels: Soft-SVM



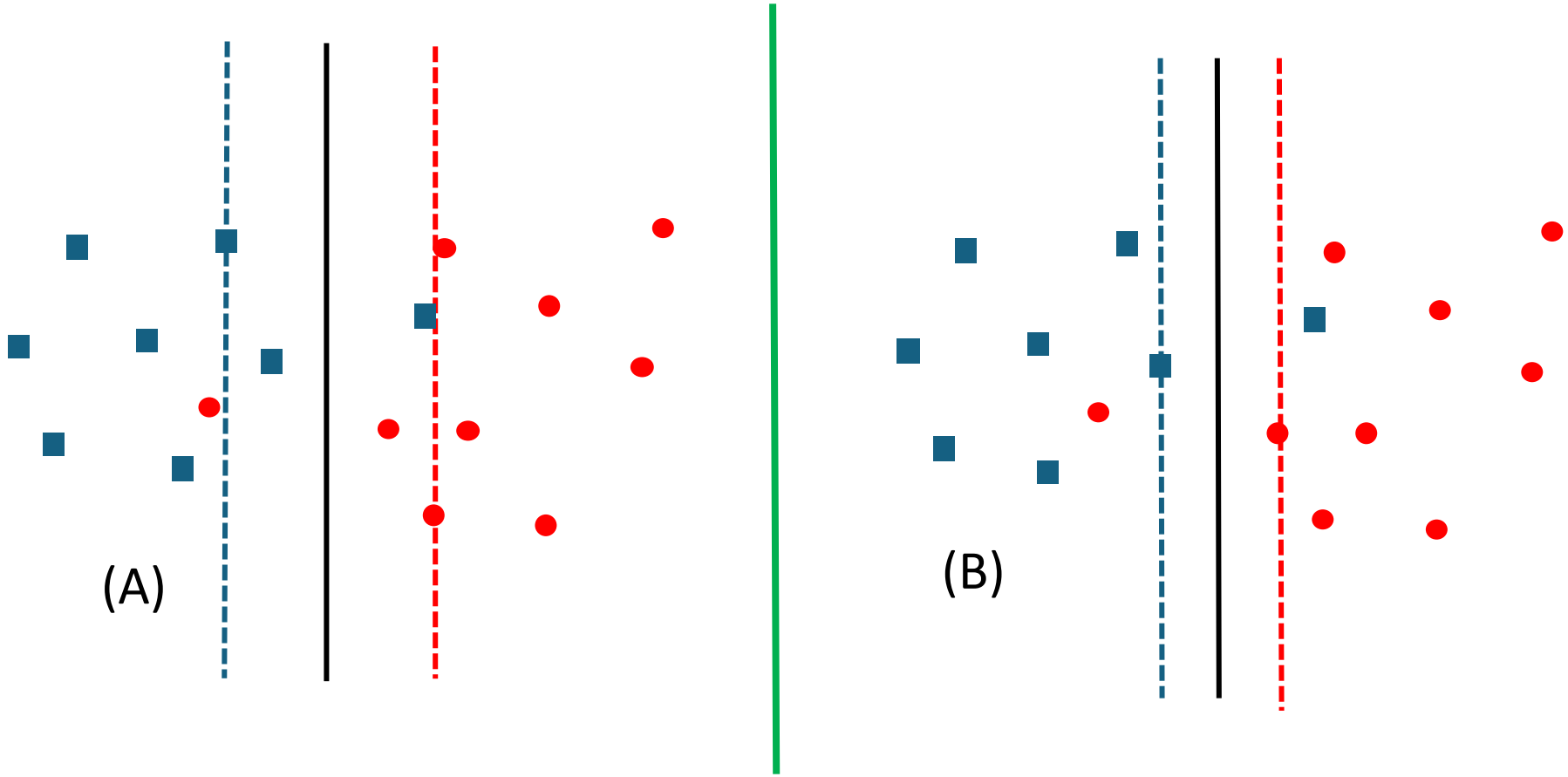
In-Class Participation Question (Q1)

SVM: not linearly separable



In-Class Participation Question (Q2)

SVM: not linearly separable.



Why Soft-Margin SVM?

- SVM objective:

$$\min_{w,b} \frac{1}{2}||w||^2$$
$$s.t.\ y_i(w^T x_i + b) \geq 1$$
$$for\ i\ \in \{1, 2, ..., N\}$$

- Limitations:
  - Data must be **linearly separable**
  - Extremely sensitive to outliers
  - Not realistic for real-world noisy datasets

- Solution: Soft-Margin SVM: allowing some mistakes:

$$\min_{w,b} \frac{1}{2}||w||^2 \ + \ [\text{allowing some mistakes}]$$

Soft-Margin SVM: Primal Formulation

- Introduce slack variables:

$$\xi_i \geq 0$$

- Optimization problem:

$$\min_{w,b,\xi} \frac{1}{2} \| w \|^2 + C \sum_{i=1}^{n} \xi_i$$

st:

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

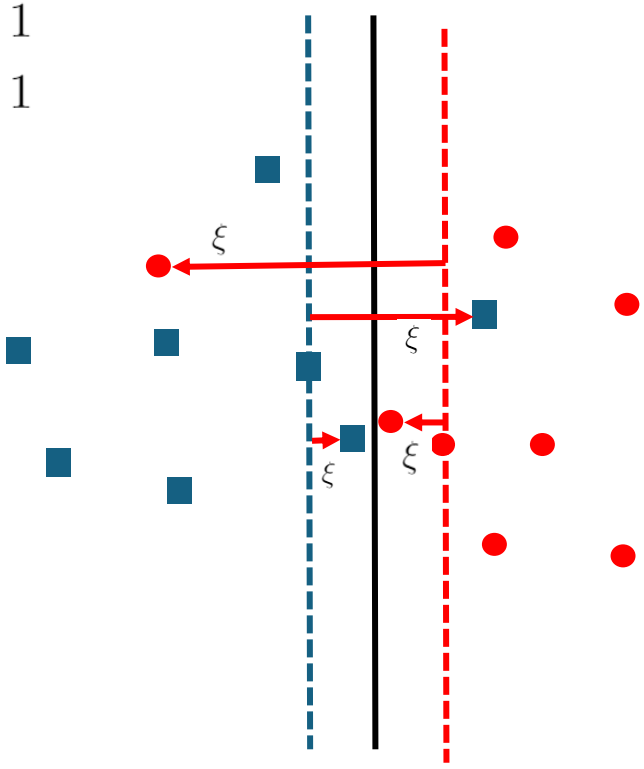
$$\xi_i \geq 0$$

Geometric Interpretation of Slack Variables

- Slack variable ξ is calculated as:

$$\xi_i = \begin{cases} 1 - y_i(w^T x_i + b) & \text{if } y_i(w^T x_i + b) < 1 \\ 0 & \text{if } y_i(w^T x_i + b) \geq 1 \end{cases}$$

For each sample:
- $\xi_i = 0$:correctly classified, outside margin
- $0 < \xi_i < 1$:inside margin but correctly classified
- $\xi_i > 1$:misclassified



- This is equivalent to the following closed form:

$$\xi_i = max\left(1 - y_i(w^T x_i + b), 0\right)$$

This is also called hinge-loss.

Soft-SVM: objective

- Objective with slack variables:

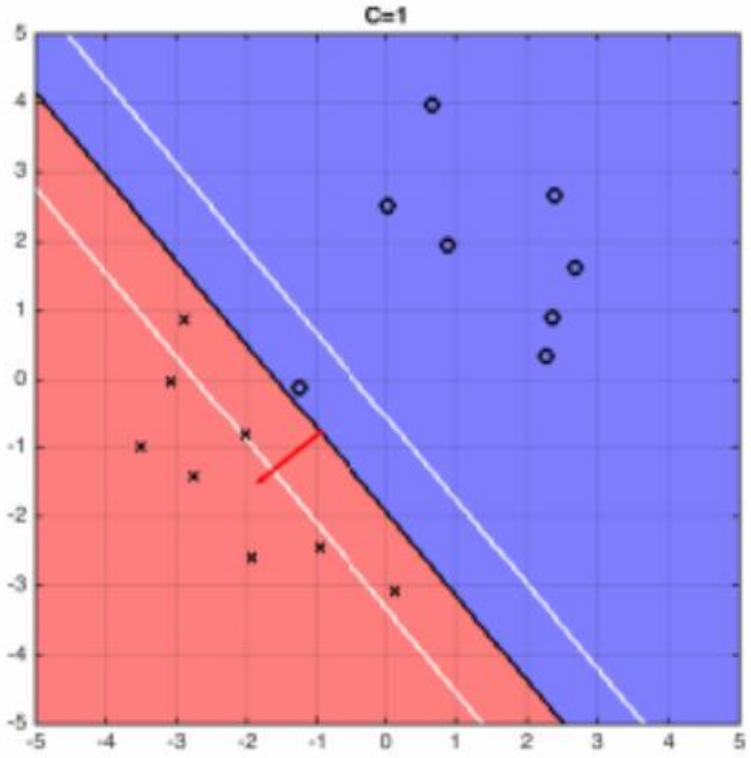
→ maximize margin → penalize violations

$$\min_{w,b} \frac{1}{2}||w||^2 + \mathcal{C}\sum_i \xi_i$$
$$s.t.\ y_i(w^T x_i + b) \geq 1 - \xi_i$$
$$\xi_i \geq 0$$
$$for\ i\ \in \{1, 2, ..., N\}$$

Soft SVM balances:
Large margin vs. classification errors

-> Allow controlled violations of the margin.

Soft-SVM: C value (large)



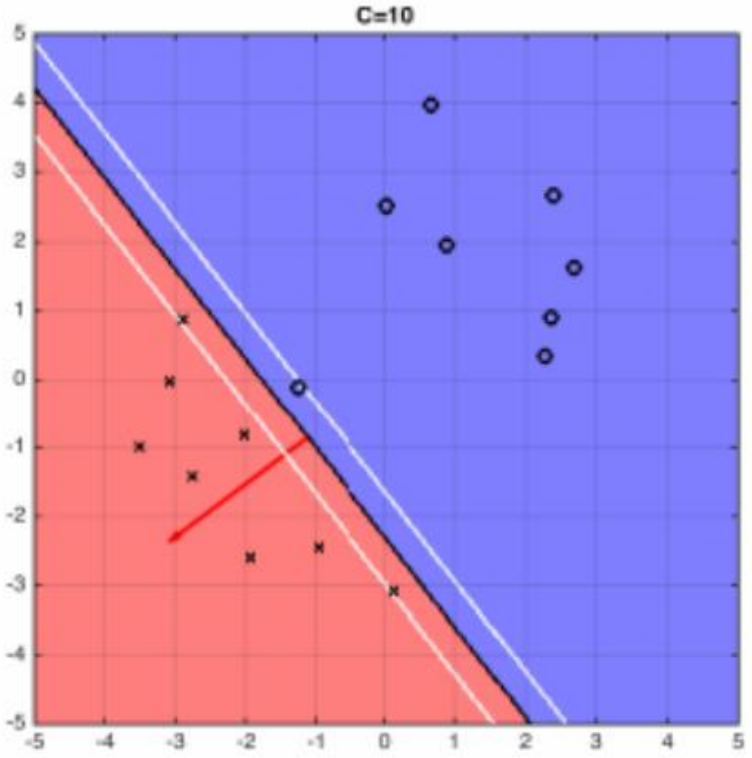
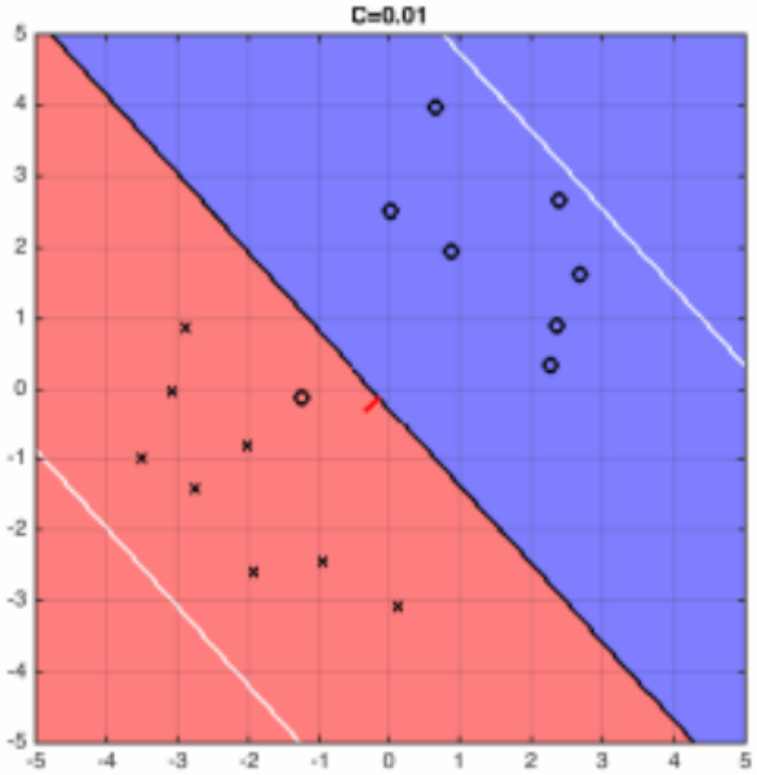


Image source: Kilian Weinberger, Cornell

Soft-SVM: C value (Small)



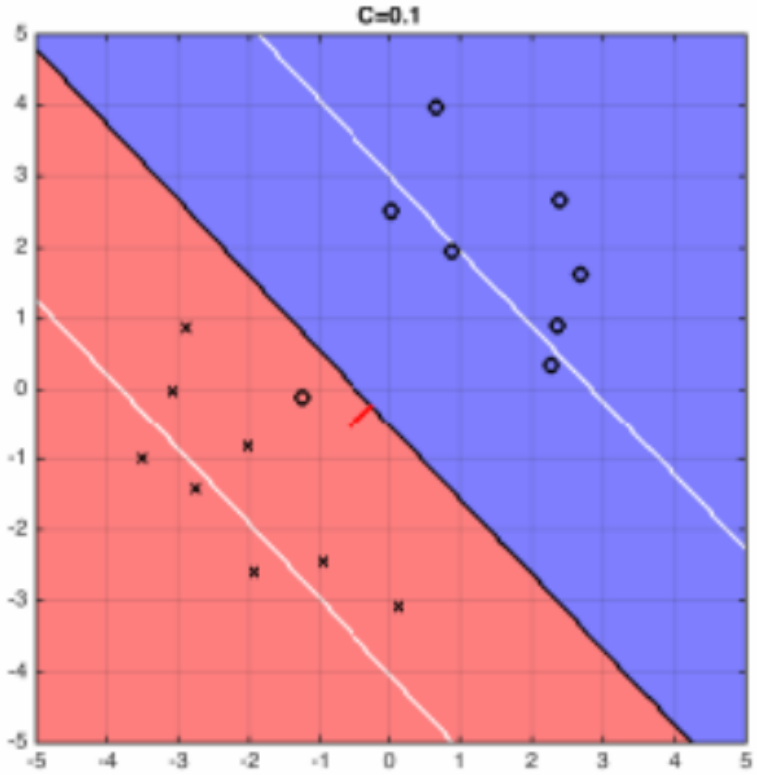


Image source: Kilian Weinberger, Cornell

Soft-SVM: hyper-parameter

- *C* increases, tendency to make mistakes decreases on training data => small margin, overfiting
- *C* decreases => large margin, underfiting

In-Class Participation Question (Q3)

$$(w^*, b^*) = \text{argmin} \frac{||w||^2}{2} + C \frac{1}{n} \sum_{i=1}^{n} \xi_i$$

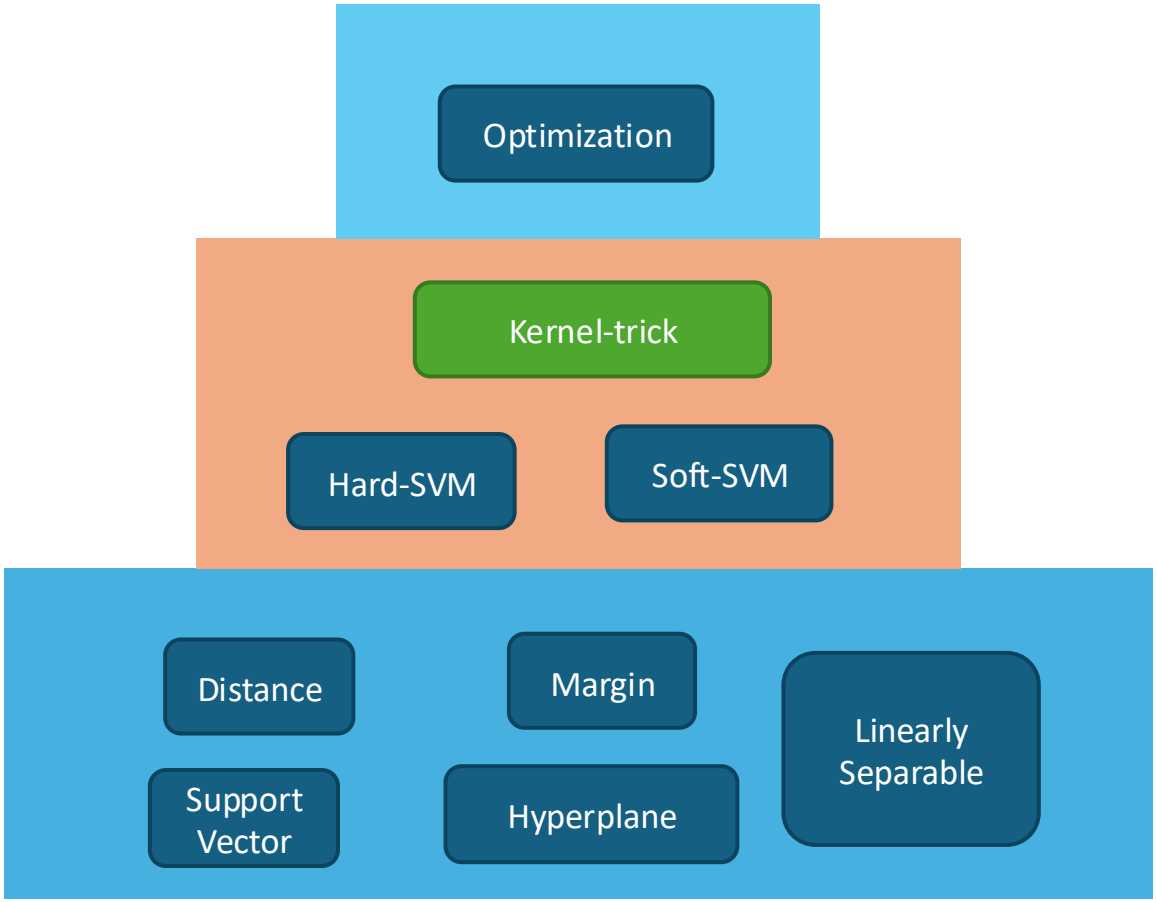
When the *C* parameter is set to infinite, which of the following holds true?

A) The optimal hyperplane if exists, will be the one that completely separates the data

B) The soft-margin classifier will separate the data

C) None of the above

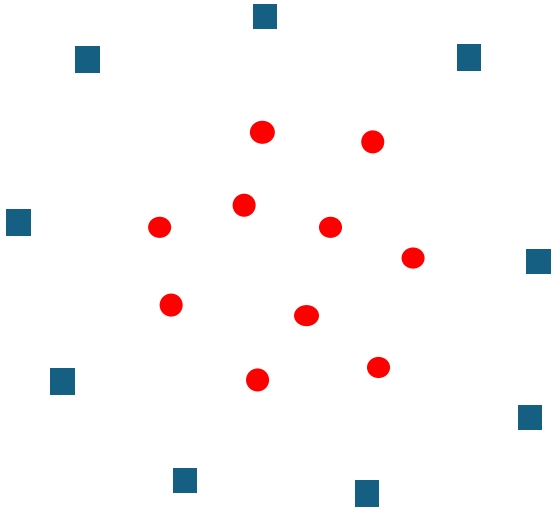
Three levels: kernel-trick



Not linearly separable (1)

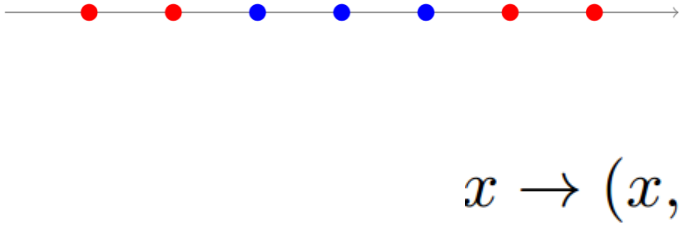
- Slack variables allow misclassified points.
- But what if data are not linearly separable



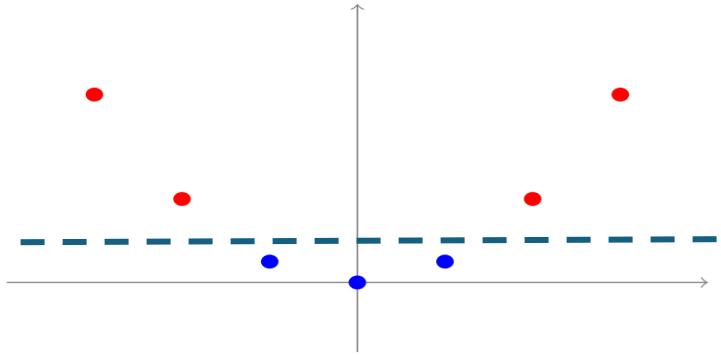


Not linearly separable (2)

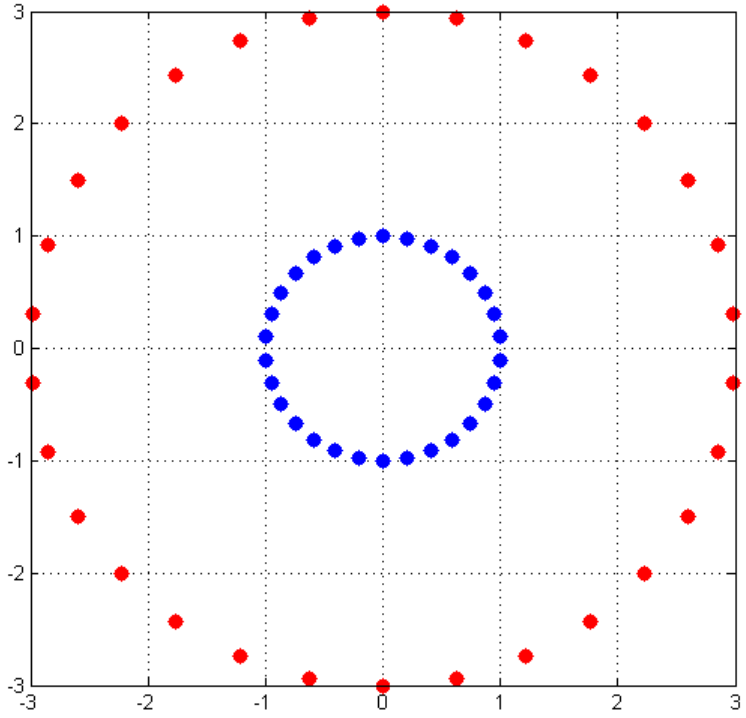
- Possible Solution: Add quadratic transformation of features



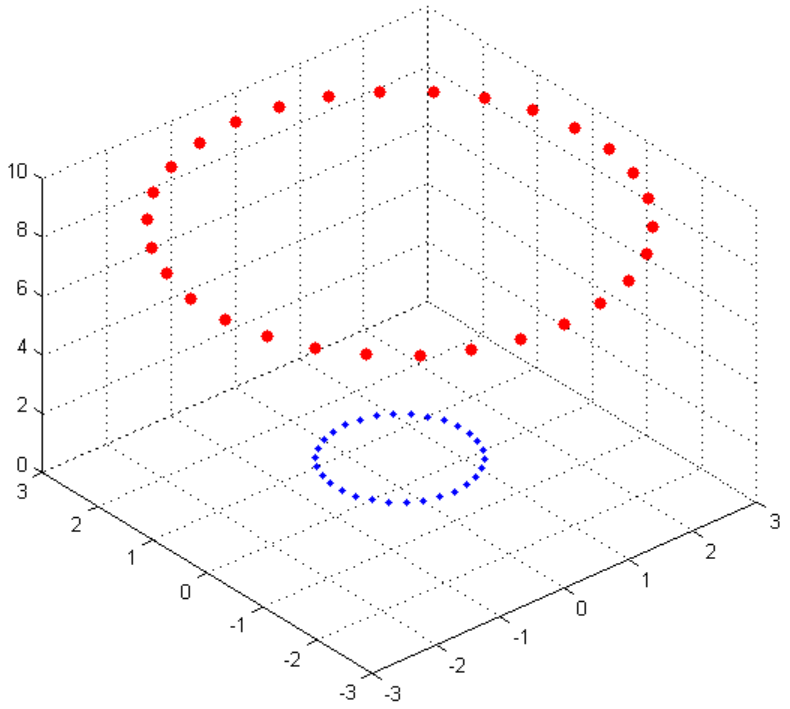
$$x \rightarrow (x, x^2)$$



Not linearly separable (3)

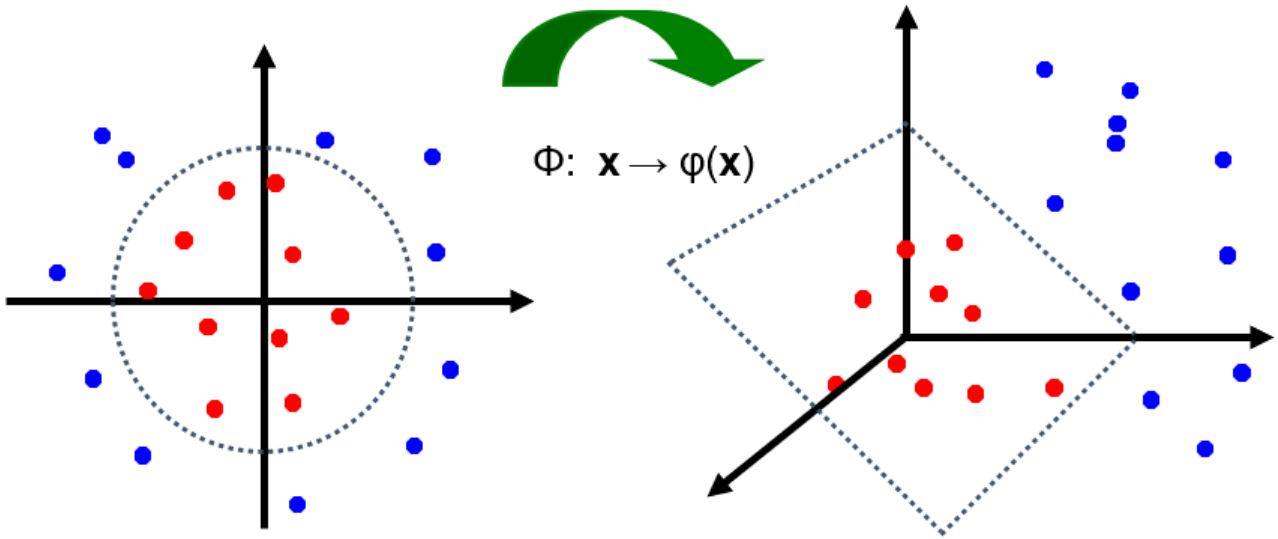


$x^2 + y^2 = 1$ $x^2 + y^2 = 9$



$(x, y) \rightarrow (x, y, x^2 + y^2)$

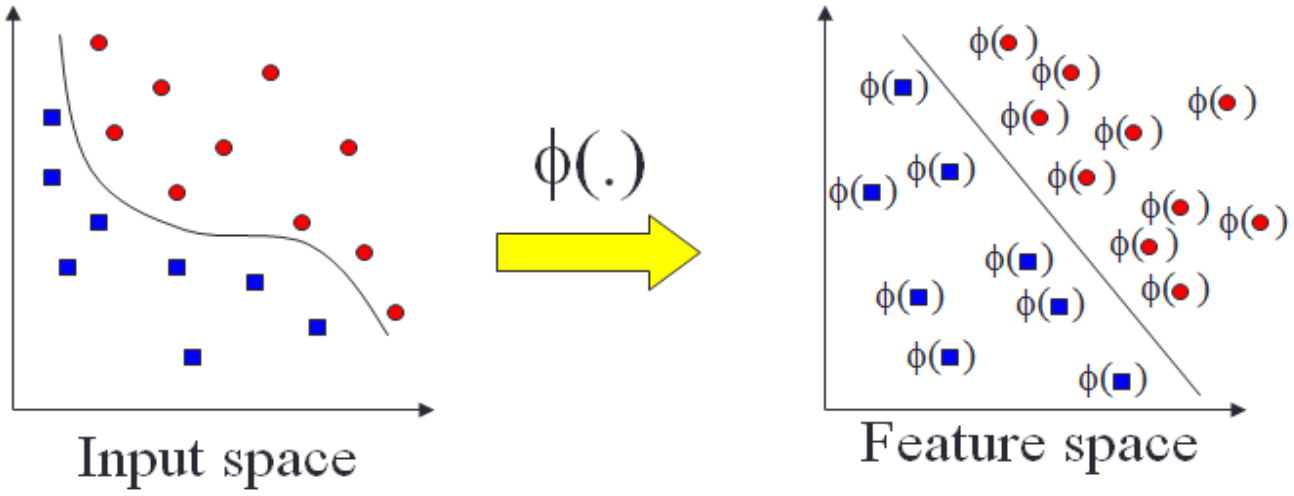
Not linearly separable (4)



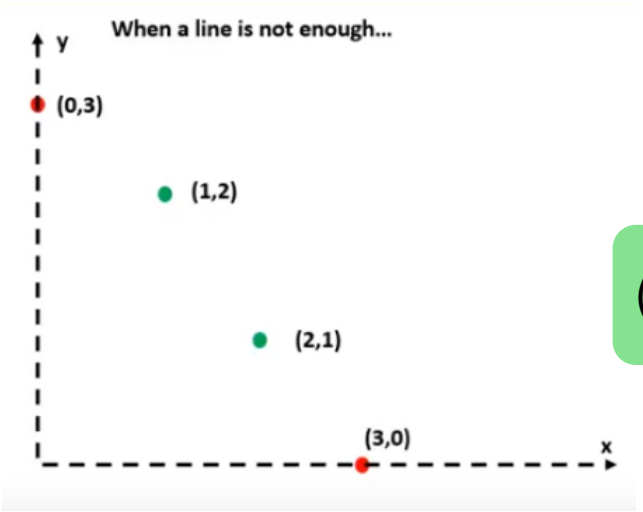
$$\Phi : \mathbb{R}^2 \to \mathbb{R}^3$$
$$(x_1, x_2) \mapsto (z_1, z_2, z_3) := (x_1^2, \sqrt{2}\, x_1 x_2, x_2^2)$$

Mapping function $\Phi(.)$

- General idea: the original input space can be mapped to some higher-dimensional feature space with mapping function $\Phi(.)$, where the training set is separable:

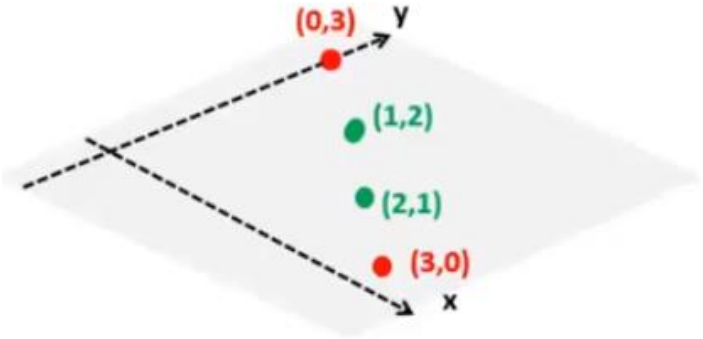


In-Class Participation Question (Q4)



(X, Y) → (X, Y, Z)

(x,y)
(0,3)
(1,2)
(2,1)
(3,0)



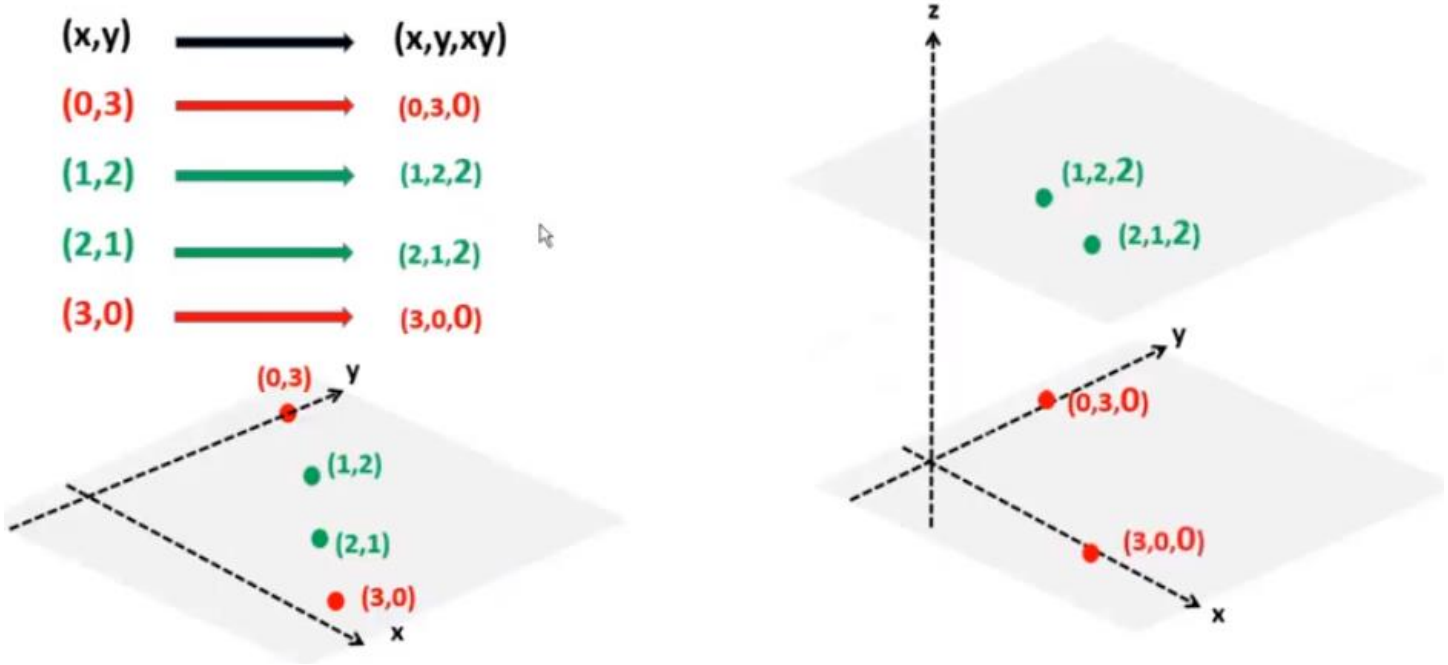
Which function should we use?

(A) $Z = X + Y$

(B) $Z = X*Y$

(C) $Z = X^2$

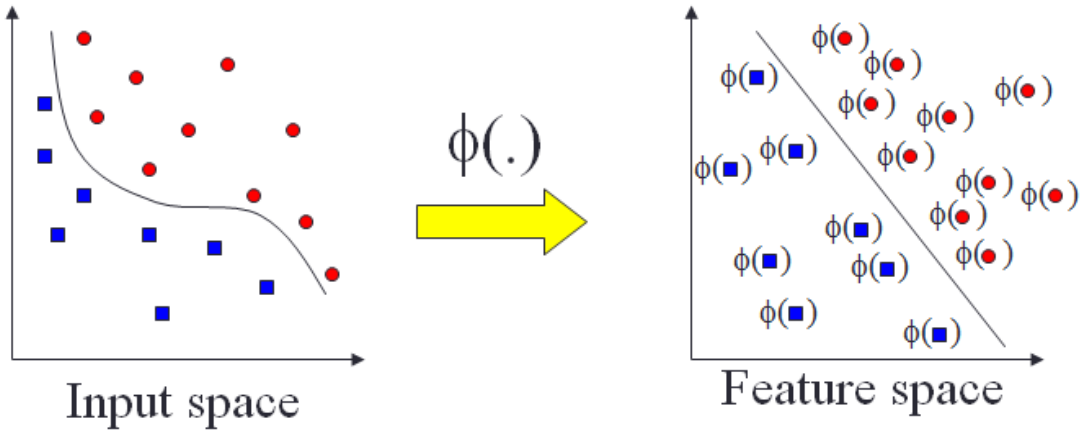
In-Class Participation Question (Q4) Ans



Issues with ϕ(x)

Mapping to a feature space can produce **problems**:

- High computational burden due to high dimensionality
- Difficult to find ϕ(x)



Kernels: for nonlinear boundaries

- *kernel function* is some function that corresponds to an inner product in some expanded feature space.

$$\mathrm{K}(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

- With a few mathematical tricks, kernels allow us to fit linear SVM classifier in higher-dimensional space, without actually computing all these features

Example: find the ϕ(x)

3-dimensional vectors $x = [x_1, x_2, x_3]^T$ $y = [y_1, y_2, y_3]^T$;

let $K(x, y) = (x^T y)^2$

What is the $\phi(x)$ that satisfy $K(x, y) = \phi(x)^T \phi(y)$?

$$K(x,y) = (x^T y)^2 = (x_1y_1 + x_2y_2 + x_3y_3)^2 = \sum_{i,j=1}^{3} x_i x_j y_i y_j$$

It is usually difficult to find such a $\phi(.)$ function

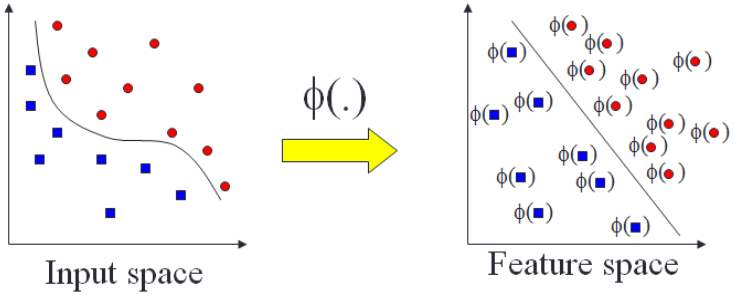
$$\phi(x) = (x_1^2, x_1x_2, x_1x_3, x_2x_1, x_2^2, x_2x_3, x_3x_1, x_3x_2, x_3^2)^T$$

$$\phi(y) = (y_1^2, y_1y_2, y_1y_3, y_2y_1, y_2^2, y_2y_3, y_3y_1, y_3y_2, y_3^2)^T$$

$$\phi(x)^T \phi(y) = \sum_{i,j=1}^{3} x_i x_j y_i y_j$$

Review: SVM optimization

- The equation:



$$\mathcal{L}(w,b,\alpha) = \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

$$0 = \sum_i \alpha_i y_i, \;\; \alpha_i \geq 0$$

- Since we have $w = \sum_i \alpha_i y_i x_i$, we have

Only depend on inner products

$$f_{w,b}(x) = w^T x + b = \sum_i \alpha_i y_i x_i^T x + b$$

Review: SVM optimization (Cont'd)

- If every data point is mapped into high-dimensional space via some transformation Φ: x → ϕ(x), the optimization equation:

$$\mathcal{L}(w,b,\alpha) = \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j \phi(x_i)^T \phi(x_j)$$

- Conclusion: no need to design ϕ(x), only need to design:

$$K(\mathbf{x_i}, \mathbf{x_j}) = \phi(\mathbf{x_i})^\top \phi(\mathbf{x_j})$$

Kernels: examples

- Polynomial:

$$K(\mathbf{x_i}, \mathbf{x_j}) = \left(1 + \mathbf{x_i}^\top \mathbf{x_j}\right)^p$$

where d is the degree of the polynomial, e.g., d = 2 for quadratic

- Gaussian:

$$K(\mathbf{x_i}, \mathbf{x_j}) = \exp\left(-\frac{\left\|\mathbf{x_i} - \mathbf{x_j}\right\|^2}{2\sigma^2}\right)$$

- Sigmoid:

$$K(\mathbf{x_i}, \mathbf{x_j}) = \tanh\left(\mathrm{c}\mathbf{x_i}^\top \mathbf{x_j} + \mathrm{h}\right)$$

Advantages: SVM

- Sparseness of solution when dealing with large data sets
  - only support vectors are used to specify the separating hyperplane
- Ability to handle large feature spaces
  - complexity does not depend on the dimensionality of the feature space
- Overfitting can be controlled by soft margin approach
- Kernels allow very flexible hypotheses, Soft-margin extension permits misclassified examples

Weakness: SVM

- It is sensitive to noise

  - A relatively small number of mislabeled examples can dramatically decrease the performance

- It only considers two classes

  - how to do multi-class classification with SVM?

    1) with m classes, learn m SVM's
       - SVM 1 learns "Output==1" vs "Output != 1"
       - SVM 2 learns "Output==2" vs "Output != 2"
       - :
       - SVM m learns "Output==m" vs "Output != m"

    2)To predict the output for a new input, just predict with each SVM and find out which one puts the prediction the furthest into the positive region.

Weakness: SVM (Cont'd)

- Especially when using kernels, not immediately clear how to interpret/extract insights
  - Choice of kernel
    - Gaussian or polynomial kernel is default
    - if ineffective, more elaborate kernels are needed
  - Choice of kernel parameters
    - e.g. σ in Gaussian kernel
  - Optimization criterion – Hard margin v.s. Soft margin
    - a lengthy series of experiments in which various parameters are tested

$$K(\mathbf{x_i}, \mathbf{x_j}) = \exp\left(-\frac{\left\|\mathbf{x_i} - \mathbf{x_j}\right\|^2}{2\sigma^2}\right)$$

After-Class Questions: (1)

Question 1: Hand-Calculating a Hard-Margin SVM on a Toy Dataset

Consider the following **toy dataset** with **three 2D data points** from two classes:

Index	$x = (x_1, x_2)$	**Label (y)**
1	(2, 2)	+1
2	(4, 4)	+1
3	(2, 0)	−1

- Plot the three points on a 2D plane.
- Write down the **primal hard-margin SVM optimization problem**.
- By geometric reasoning, determine:
  - the **maximum-margin separating hyperplane**,
  - the corresponding **weight vector w**and **bias** b.
- Identify which points are **support vectors** and explain why.

Hint: The optimal hyperplane should maximize the distance between the closest points from both classes.

After-Class Questions: (2)

Soft-Margin SVM and the Role of C:

- Now consider a **non-separable dataset** generated using Python.
- Use sklearn.datasets.make_classification to randomly generate a **2D dataset** with overlapping classes.
- Train a **linear soft-margin SVM** using sklearn.svm.SVC(kernel="linear").
- Experiment with different values of C:

$$C \in \{0.01, 0.1, 1, 10, 100\}$$

- For each value of C:
  - plot the decision boundary,
  - report the number of support vectors,
  - comment on margin width and misclassification behavior.
- **Discussion questions:**
  - What happens when $C \to \infty$?
  - What happens when $C \to 0$?
  - How does C control the tradeoff between **margin maximization** and **training error**?